



GE VERNOVA

ADMS

16.25.1

Enterprise OMS Installation Guide

Copyright

Copyright 2026, GE Vernova and/or its affiliates ("GE Vernova"). All Rights Reserved.

GE and the GE Monogram are trademarks of General Electric Company used under trademark license.

This document is the confidential and proprietary information of GE Vernova and may not be reproduced, transmitted, stored, or copied in whole or in part, or used to furnish information to others, without the prior written permission of GE Vernova.

Change History

Date	Change Description
Jul 3, 2023	ADMS 3.16: • Updated Software Prerequisites. • Updated Windows Configuration. • Updated Parameters for work-model-service. • Updated Parameters for work-model-eventhandler. • Rebranded title page to ADMS as EOMS is part of ADMS. • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • Updated Kubernetes Server. • Updated Software Prerequisites. • Updated Generating Certificates. • Updated Generating Certificates. • Updated Installing Prerequisites. • Updated Parameters for work-model-service. • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Mar 20, 2024	ADMS 16.7.0: • Updated document format and styling.
May 9, 2024	ADMS 16.8.0: • Updated Parameters for work-model-eventhandler.

Contents

1. Introduction	6
2. Installation Overview	8
2.1. Software Delivery Media	8
2.2. Installing in Kubernetes	9
2.2.1. Hardware and Operating System Prerequisites	9
2.2.2. Software Prerequisites	10
2.2.2.1. Kubernetes Server	10
2.2.2.2. Deployment Server	10
2.3. Installing in Windows	10
2.3.1. Hardware and Operating System Prerequisites	11
2.3.2. Software Prerequisites	11
3. Configuring Enterprise OMS	12
3.1. Preparing for Installing in Kubernetes	12
3.1.1. Setting Up the Infrastructure Layer	12
3.1.2. Installing the Helm and ArgoCD based deployment Tool	12
3.1.3. Air-Gapped Deployment	12
3.1.4. Configuring Foundation Base Components	12
3.1.5. Preparing Setup Folders	13
3.1.6. Generating Certificates	13
3.1.7. Configuring the YAML File	17
3.1.8. Helm Based Deployment	18
3.1.9. ArgoCD Based Deployment	19
3.1.10. Setting Up Firewall Rules	26
3.2. Preparing for the Installing in Windows	26
3.2.1. Generating Certificates	26
3.2.2. Setting Up Firewall Rules	29
3.3. Configuring EOMS Components for a Multi-Host System	30
3.3.1. Windows Configuration	30
3.3.2. Kubernetes Configuration	31
4. Configuring ADMS	33
4.1. Configuring the hosts.etc File	33
4.2. Importing Certificates to the ADMS Hosts	33
4.3. Configuring ADMS Components	39
4.4. Configuring ADMS Components for a Multi-Host System	45
5. Installing Enterprise OMS	48
5.1. Installing Prerequisites	48
5.2. Installing in Kubernetes	50
5.3. Installing in Windows	51
5.3.1. Setting Up the Database	52
5.3.2. Configuring Database SSL Configuration	54

5.3.3. Upgrading Work Model Database to a Newer Version	55
5.3.4. Upgrading PostgreSQL to a Newer Version	55
6. Advanced Configuration	57
6.1. Configuring Enterprise OMS Services	57
6.1.1. Parameters for Every Service	57
6.1.2. Parameters for work-model-notifications	58
6.1.3. Parameters for work-model-eventhandler	58
7. Running Enterprise OMS	60
7.1. Running ADMS Interface Servers	60
7.2. Running Enterprise OMS Services in Kubernetes	61
7.3. Running Enterprise OMS Services in Windows	61
8. Maintenance and Troubleshooting	64
8.1. Verifying the Setup	64
8.2. Upgrading to a Newer Version	64
8.3. Maintenance	64
8.3.1. Database Schema	64
8.3.2. Profiling	64
8.3.3. Services	64
8.3.4. Endpoints	65
9. Generating an SSL/TLS Certificate	66
9.1. Required Tools	66
9.2. Self-Signed Certificate Authority	66
9.3. Create Service Certificate	67
9.4. Add Root CA to Trust Store	69
9.5. Additional Certificate Operations	70
9.5.1. Common Keytool and Openssl Commands	70
10. Enterprise OMS Journaling and Archiving	72
11. Enterprise OMS Organizations Definition	74
11.1. Determining Organization Types and Organizations	75
11.2. Defining the Criteria for Assigning Organizations	76
11.3. Define a Default "Fall-Back" Organization Name	77
11.4. Defining Organization Types and Organizations	78
11.5. Using the Permit Application to Define Organizations	79
11.5.1. Adding AREAs	80
11.5.2. Adding MODEs	81
11.5.3. Granting AREA Access to MODEs	82
11.5.4. Adding User Types	85
11.5.5. Assigning MODEs to User Types	86
11.5.6. Defining Users	88
11.5.7. Validating and Saving the Model	89
11.6. Defining Criteria for Assigning Organizations	89
11.7. Defining ESRI Shape Files	90

11.8. Applying ESCRI Shape Files to the Customer Model	92
12. Configuration Files	94
12.1. WorkOrderAdapterConfig.xml	94
12.2. DataCorrectionInterfaceConfig.xml	95

1. Introduction

Enterprise OMS is a GE solution to support detailed management of work on the electrical network. Orders are used to facilitate tracking and managing work for repairing or maintaining network equipment. Orders can include multiple tasks with different objectives to accurately identify and report the required work.

Currently, the Enterprise OMS system integrates with GE's control room distribution platform, Advanced Distribution Management System (ADMS).

The Enterprise OMS solution uses multiple services that support the order management, database, communication with ADMS, and security aspects. Specifically:

- The Enterprise OMS user interface is integrated into the ADMS Outage Management System (OMS).
- The Work Order Adapter application supports communication with ADMS.
- The Enterprise OMS database in PostgreSQL stores Enterprise OMS data.
- The Enterprise OMS security features support an HTTPS connection with TLS termination.

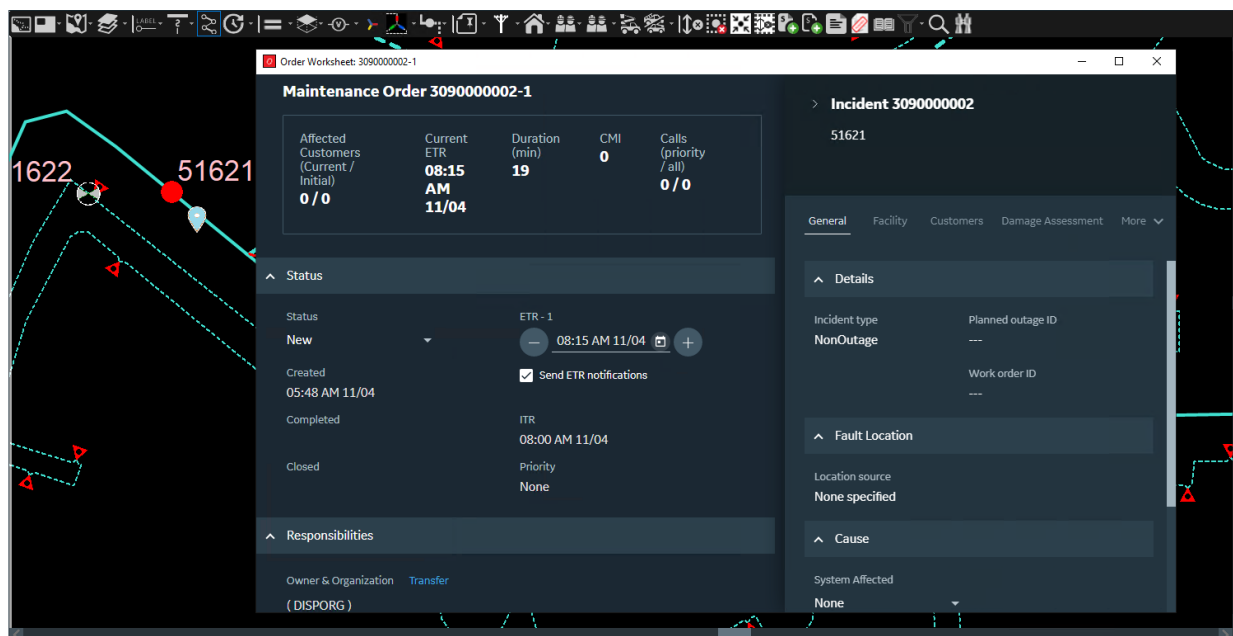


Figure 1. Enterprise OMS - Updating an Order

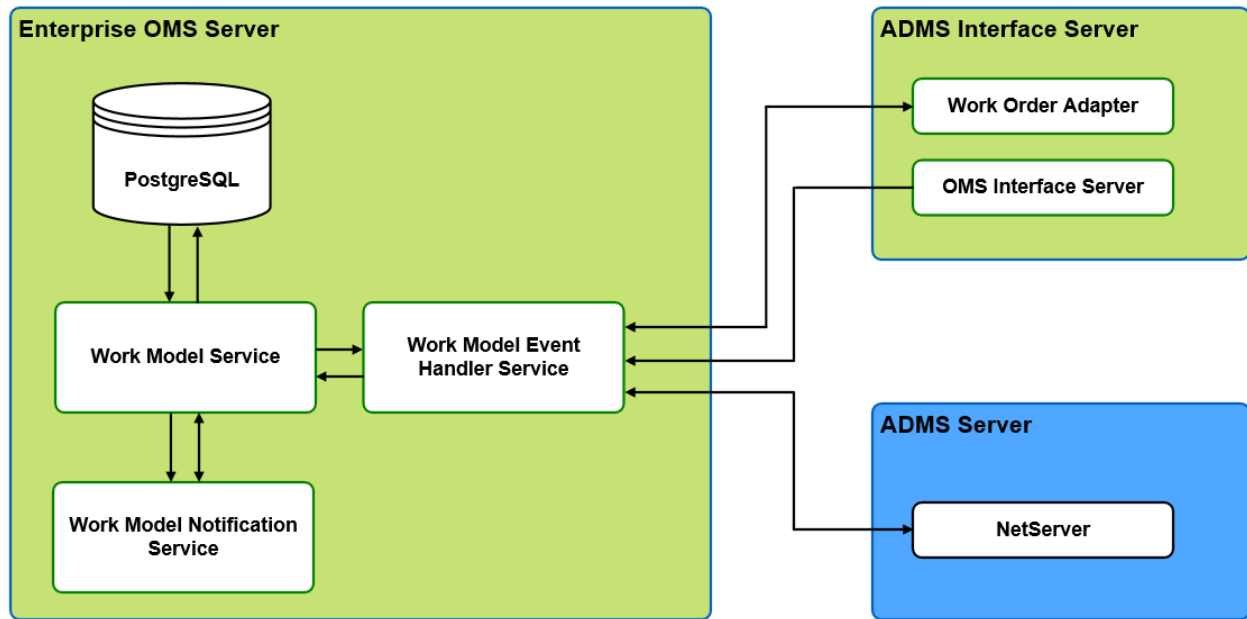


Figure 2. Enterprise OMS Process

Figure below shows tables in the Enterprise OMS database in PostgreSQL.

Tables (15)
> associate_orders
> crew_assignments
> device_operations
> locks
> order_customfield_definitions
> order_customfields
> order_events
> order_task_type
> order_type
> ordergroup_counter
> orders
> properties
> task_result
> task_type
> tasks

Figure 3. Enterprise OMS Database

2. Installation Overview

The Enterprise OMS system supports two setup options:

- Installing and running under Kubernetes.
- Installation using a Windows installer and running under Habitat process manager (PROCMAN).

2.1. Software Delivery Media

Enterprise OMS components are delivered with the ADMS Interfaces kit, solution artifacts, and configuration files:

- **ADMS_interfaces.msi:** Includes ADMS interface servers and Enterprise OMS components (Work Order Adapter, Work Order Archive Adapter, and Work Order Database Sync applications).
- **ADMS_server.msi:** Includes ADMS servers, including NetServer (OMS Server).
- **ADMS_client.msi:** Includes the ADMS client component, including OMS Display Plugin.
- **ADMS_tools.msi:** Includes ADMS tools, including OMS Client Configuration Editor.

For installing in Kubernetes:

- **foundation helm chart :** Defines the deployment settings for Foundation Base using the helm.
- **work-model helm chart:** Defines the deployment settings for Enterprise OMS using the helm.

A sample folder structure that is used for deployment is shown below. This structure can be used as a reference to create the folder structure.

```
├── k8s-base-manifest.yaml
├── Foundation helm chart
├── Work Model helm chart
├── apps
│   ├── Custom-namespace.yaml
│   ├── foundation-base.yaml
│   ├── foundation-osb.yaml
│   ├── foundation-ui-server.yaml
│   ├── foundation-data.yaml
│   └── work-model.yaml
├── apps-config
└── work-model-values.yaml
```

- **foundation-base-cache-<TAG>.tar.gz:** Cache container image with artifacts needed for Foundation Base deployment. Used for deployment in environments without connection to the GE Docker registry (DTR).
- **work-model-cache-3.16.0.0-7.tar.gz:** Cache container image with artifacts needed for Enterprise OMS deployment. Used for deployment in environments without connection to the GE Docker registry (DTR).

For installing using a Windows installer:

- **ADMS_Enterprise_OMS_Module.msi:** Installer for the Enterprise OMS deployment.

2.2. Installing in Kubernetes

Enterprise OMS components (containers) are orchestrated by Kubernetes. Therefore, you must first create a Kubernetes environment before deploying Enterprise OMS. The application deployment is certified for a "vanilla" (open source) Kubernetes cluster.

The Enterprise OMS deployment is based on the Bsfdeploy client, which deploys the Enterprise OMS helm chart to the Kubernetes cluster.

The Enterprise OMS deployment in Kubernetes requires the security component of GE's common Kubernetes-based platform (hereinafter referred to as "Foundation").

Helm helps you define, install, and upgrade Kubernetes applications and manage Helm charts.

2.2.1. Hardware and Operating System Prerequisites

Enterprise OMS has the following hardware requirements:

- **Kubernetes Server:** A high availability topology-capable (HA) cluster setup is highly recommended. Use at least one node with the following properties:
 - **Memory:** 16 GB per node.
 - **CPU:** 4 cores per node.
 - **Storage:** At least 20 GB per node.
 - **Operating System:** Red Hat Enterprise Linux 7.5 or later, or CentOS 7.x.
- **Deployment Server:** A server specifically for deployment purposes.
 - **Memory:** 16 GB.
 - **CPU:** 4 cores.
 - **Storage:** 200 GB.
 - **Operating System:** Microsoft Windows Server 2016, Windows Server 2019, Windows 10, Red Hat Enterprise Linux 7.5, or CentOS 7.x.
- **Clients:** The Enterprise OMS user interface is embedded into the ADMS client. The following requirements apply to ADMS clients.
 - **Memory:** 16 GB.
 - **CPU:** 2 @ 2.27 GHz processors with 4 cores each.
 - **Storage:** 5 GB.
 - **Monitor:** A monitor with full HD (1920x1080) display resolution is recommended.
 -

Display: Due to the demanding nature of the displays of the distribution networks, higher-performance video cards that support multiple high-resolution monitors are recommended.

- **Scaling:** The certified display scaling for text, applications, and other items on the screen is 100%. Display layout issues may appear at higher size scaling.
- **Resolution:** The certified display resolution is 1920x1080. Ultra high display resolutions (for example, 3840 x 2160) are not certified. Display layout issues may appear.
- **Operating System:** Microsoft Windows 10.

For ADMS server and client requirements, refer to the *ADMS Release Notes*.

2.2.2. Software Prerequisites

2.2.2.1. Kubernetes Server

The Enterprise OMS server has the following software requirements:

- Red Hat Docker version docker-1.13.1-91 or later (for deployment in Red Hat Linux)
- CentOS Docker version docker-1.13.1-91 or later (for deployment in CentOS)
- Kubernetes version 1.13.3 or later
- Helm version 3 or later
- Docker Trusted Registry or a similar supported API/v2 registry (for example, Nexus)
- Amazon Corretto 17.0.8.1
- PostgreSQL 12.5

2.2.2.2. Deployment Server

The Enterprise OMS deployment server has the following software requirements:

- Docker CE version 18.06.1 or later
- Helm version 3 or later (optionally)
- Docker Trusted Registry or a similar supported API/v2 registry (for example, Nexus)

2.3. Installing in Windows

Enterprise OMS services can also be installed using a Windows installer, `ADMS_Enterprise_OMS_Module.msi`.

2.3.1. Hardware and Operating System Prerequisites

Enterprise OMS has the following minimum hardware requirements:

- **Enterprise OMS Server:**
 - **CPU:** 4 cores
 - **Memory:** 16 GB of RAM
 - **Operating System:** Windows Server 2019 or Windows Server 2016.
- **Clients:** The Enterprise OMS user interface is embedded in the ADMS client. The following requirements apply to ADMS clients.
 - **Memory:** 16 GB.
 - **CPU:** 2 @ 2.27 GHz processors with 4 cores each.
 - **Storage:** 5 GB.
 - **Display:** Due to the demanding nature of the displays of the distribution networks, higher-performance video cards that support multiple high-resolution monitors are recommended.
 - **Monitor:** A monitor with full HD (1920x1080) display resolution is recommended.
 - **Scaling:** The certified display scaling for text, applications, and other items on the screen is 100%. Display layout issues may appear at higher size scaling.
 - **Resolution:** The certified display resolution is 1920x1080. Ultra high display resolutions (for example, 3840 x 2160) are not certified. Display layout issues may appear.
 - **Operating System:** Microsoft Windows 10.

For ADMS server and client requirements, refer to the *ADMS Release Notes*.

2.3.2. Software Prerequisites

- **Server**
 - Amazon Corretto 17.0.8.8.1
 - PostgreSQL 15

3. Configuring Enterprise OMS

The Enterprise OMS setup process includes the configuring the Enterprise OMS and ADMS components.

3.1. Preparing for Installing in Kubernetes

3.1.1. Setting Up the Infrastructure Layer

For more information about setting up the Kubernetes cluster, refer to the "Setting Up the Infrastructure Layer" section in the *Foundation Installation Guide*.

3.1.2. Installing the Helm and ArgoCD based deployment Tool

The deployment using Helm chart of foundation and EOMS. For installation and usage details, refer to the *Foundation Documentation*.

3.1.3. Air-Gapped Deployment

Deployment into air-gapped environments is supported. Air-gap artifacts (available from Digital Grid Artifacts) must be made available on the deployment server where you run the helm-chart, but the only requirement for the targeted nodes is SSH access.

For the deployment process, refer to the "Air-Gap Support" section in the *Foundation Installation Guide*.

For details about setting up a deployment server, refer to the "Requirements for Deployment Server" section in the *Foundation Installation Guide*.

3.1.4. Configuring Foundation Base Components

The Enterprise OMS solution uses Foundation Base, which includes the security components of Foundation. For more information about Foundation, consult your GE project team.

The Foundation Base services use cryptographic keys and certificates in these areas:

- The Ingress Controller to perform TLS termination.
- Public/Private key pair for UAA to sign access tokens.
- Trusted Certificates for LDAPS clients to validate the authenticity of the LDAPS server. (You can skip this step if your LDAPS server uses a certificate signed by a trusted certificate authority.)

These keys and certificates must be created *prior to* deploying Foundation Base.

3.1.5. Preparing Setup Folders

! Important

Skip this section if you set up the system using the Windows installer.

To prepare the setup folders:

1. Create a local folder for deployment (referred to as <App_Folder> in the remainder of this document). You will deploy the Enterprise OMS application from this folder.
2. Copy the foundation-base.yaml and wm-manifest.yaml files to <Enterprise_OMS_Configuration_Folder>.
3. Copy the admin.conf file that your Kubernetes instance uses from /etc/kubernetes to <Enterprise_OMS_Configuration_Folder>.
4. Copy the root certificate authority and Ingress Controller certificates content in YAML file. For details about generating the key and certificate, see [Generating Certificates](#).

A sample folder structure that is used for deployment is shown below. This structure can be used as a reference to create the folder structure.

```
├── k8s-base-manifest.yaml
├── Foundation helm chart
├── Work model helm chart
├── apps
│   ├── Custom-namespace.yaml
│   ├── foundation-base.yaml
│   └── work-model.yaml
├── apps-config
└── work-model-values.yaml
```

In the example above, the deployment folder contains all the required files for deploying the Foundation Base and Enterprise OMS helm-chart's.

3.1.6. Generating Certificates

Secure connections between the Enterprise OMS and the ADMS system is established using an HTTPS connection with TLS termination.

For HTTPS with TLS, you need to obtain SSL certificates from your certificate authority. The certification authority is based on Active Directory or other services. Therefore, you need to have two certificates: one for Kubernetes Ingress Controller and another for the ADMS interface server.

For Enterprise OMS, you need the following certification files:

- Root certificate authority certificate. This certificate is used to issue and sign certificates. (For example, file name = ca.p12; CN = Work Model Root CA).
- Trust Store file (cacerts). The cacerts file is needed for HTTPS when Enterprise OMS services run

in Java. Java services read certificates from this file instead of the local system storage (for example, storage that is configured by the Microsoft Management Console). The cacerts file is delivered with the Java Runtime Environment and the Java Development Kit. It is typically located at `JAVA_HOME/lib/security/cacerts`. This file contains all the standard Trusted Certificate Authorities for the internet.

You can copy cacerts, and then add the Certificate Authority that is used to sign all the Enterprise OMS certificates. If a trusted Certificate Authority is used, then only a copy of the cacerts is required. If a Self-Signed Certificate Authority is created, then that Self-Signed Certificate Authority must be added to the cacerts file that was copied.

- Ingress Controller certificate and key. These files are used for establishing connections with Active Directory. (For example, file name = `ingress-local-tls.crt` and `ingress-local-tls.key`; `ingress.local`)

If the Enterprise OMS system is deployed in Kubernetes with an Ingress Controller, the Ingress Controller needs a certificate that is signed by the same certificate authority that has signed the other Enterprise OMS certificates. It is recommended to have a separate certificate for the Ingress Controller, but you can use the Work Model Event Handler certificate if it contains the correct DNS hostname in the Subject Alternative Name section.

- ADMS server certificate. This certificate is used for secure communication between the Kubernetes cluster and ADMS interface servers. (For example, file name = `keystore.p12`; CN = IDMS)
- PostgreSQL certificate. This certificate is used for establishing secure connections with the PostgreSQL database. (For example, file name = `postgres.pem`; CN=postgres)

If SSL is enabled for PostgreSQL, you need the root certificate authority certificate (`ca.p12`) and ADMS server certificate and key (`keystore.p12`) on the ADMS Interface Server host. You also need the PostgreSQL certificate (`postgres.p12`) on the Enterprise OMS host to enable the Work Model Event Handler service to connect to the PostgreSQL server.

For more information about configuring PostgreSQL SSL, see [Configuring Database SSL Configuration](#).

For production, it is recommended to use certificates issued by a certificate authority. For testing and demo purposes, you can use self-signed certificates.

The example below shows how to generate OpenSSL certificates using the `genrsa` tool. This is a standard tool for most Linux distributions. For more information, see <https://www.openssl.org/source/>. The PostgreSQL certificate is generated using the "keytool" application, which is part of the Java Development Kit (JDK) 17.

For more information about generating and using certificates, see [Generating an SSL/TLS Certificate](#).

To generate a self-signed certificates:

1. Generate a root certificate authority certificate and key:

```
openssl genrsa -out rootCA.key 2048
openssl req -x509 -new -nodes -key rootCA.key -sha256 -days 3650 -out rootCA.crt -subj
"/C=US/ST=FL/O=GE Digital/CN=Work Model CA"
keytool -import -trustcacerts -keystore "%JAVA_HOME%\lib\security\cacerts" -storepass changeit
-alias WorkModelCA -file rootCA.crt
```

2. Add the Self-Signed Certificate Authority to the cacerts file:

```
copy %JAVA_HOME%\lib\security\cacerts .
keytool -import -trustcacerts -keystore .\cacerts -storepass changeit -alias WorkModelCA -file
rootCA.crt
```

3. Generate a certificate and key for Ingress Controller:

Note

It is recommended to set CN to the actual Ingress Controller DNS host name (web address you access when connecting to a secure site).

```
openssl genrsa -out ingress-local-tls.key 2048
openssl req -new -key ingress-local-tls.key -out ingress-local-tls.csr
openssl x509 -req -in ingress-local-tls.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out
ingress-local-tls.crt -days 5000
```

4. Generate a certificate and key for the Enterprise OMS services:

Note

It is recommended to set CN to the actual ADMS DNS host name (web address you access when connecting to a secure site).

```
openssl genrsa -out service.key 2048
openssl req -new -sha256 -key service.key -subj "/C=US/ST=CA/O=GE Digital/CN=localhost" -out
service.csr
openssl x509 -req -in service.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out
service.crt -days 3650 -sha256
openssl pkcs12 -export -nodes -out keystore.p12 -inkey service.key -in service.crt -certfile
rootCA.crt -name service -passout pass:
```

5. Generate a certificate for PostgreSQL:

The certificate generated for connecting to the PostgreSQL database is like the certificates generated for the other Enterprise OMS services with one exception. The Common Name (CN) for the Certificate must be the same as POSTGRES_USER. For example, if the username to connect to the database is called "postgres", the CN value for the certificate must be CN=postgres.

```
keytool -genkeypair -alias postgres -dname "CN=postgres" -validity 3650 -keyalg RSA -keysize 2048
-keystore postgres.p12 -storetype PKCS12 -keypass password -storepass password
```

```
keytool -keystore postgres.p12 -storepass password -certreq -alias postgres | keytool -keystore
ca.p12 -storepass password -gencert -alias ca -ext ku=c:dig,keyEnc -ext eku=sa,ca -rfc >
postgres.pem
keytool -keystore postgres.p12 -noprompt -storepass password -importcert -trustcacerts -alias ca
-file ca.pem
keytool -keystore postgres.p12 -noprompt -storepass password -importcert -alias postgres -file
postgres.pem
# Install the client keypair into the work-model-service
mv postgres.p12 ./work-model-service/postgres.p12
```

6. Export the server.crt, postgres.crt certificates, and the server.key key from the PKCS12 keystore containers:

```
keytool -exportcert -alias service -keystore postgres.p12 -storepass password -rfc -file
postgres.crt
keytool -exportcert -alias service -keystore keystore.p12 -storepass password -rfc -file
server.crt
openssl pkcs12 -in keystore.p12 -nodes -nocerts -out server.key
```

7. When installing in Kubernetes, copy the Trust Store file and the root certificate authority certificate, ADMS server certificates, Ingress Controller certificates, and PostGRE SQL certificate to <Enterprise_OMS_Configuration_Folder>\cert.
8. When installing in Windows and using a Self-Signed Certificate Authority, copy the Trust Store file (cacerts) and the ADMS server certificate to a local folder.

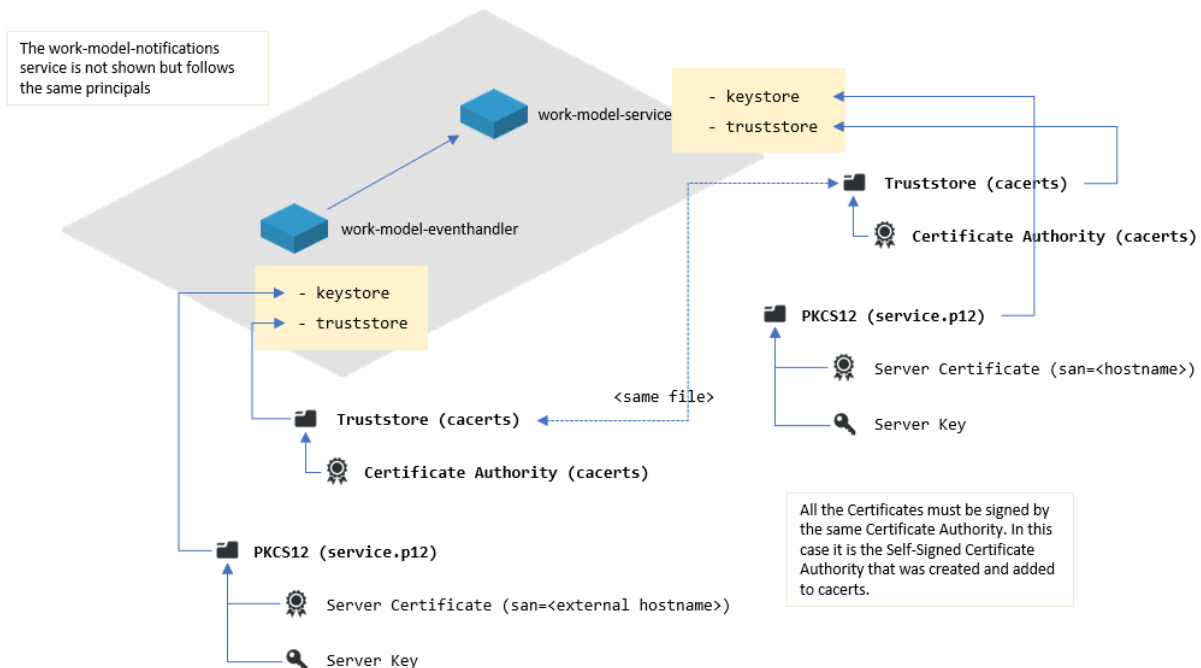


Figure 4. Certificate Flow for Enterprise OMS Services

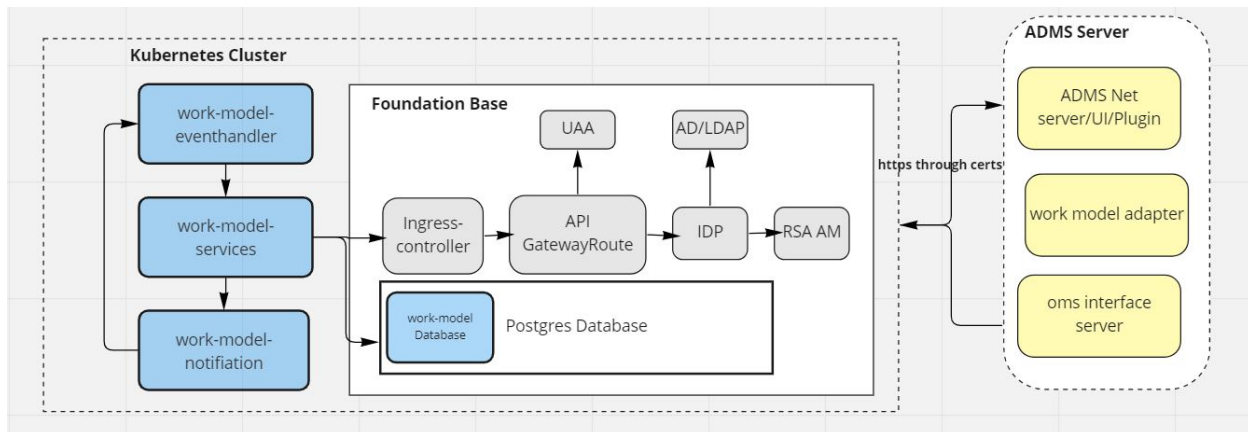


Figure 5. Certificate Flow for the Enterprise OMS System in Kubernetes

3.1.7. Configuring the YAML File

! Important

Skip this section if you set up the system using the Windows installer.

The YAML file `Values.yaml` defines both application settings and deployment settings for the Enterprise OMS components.

To configure the `custom-namespace.yaml`, `foundation-manifest.yaml` and `wm-manifest.yaml` files in `<Enterprise_OMS_Configuration_Folder>`:

1. Update the custom namespace yaml file by adding work model namespace as **work-model** and disable the `istio_injected` flag
2. Configure IP addresses according to your current configuration.
3. Set "image" properties to the paths to YAML in the Docker Registry (GE's DTR or a Nexus registry on a deployment server). When deploying from Nexus, prefix the path with the Nexus host and port.
4. When deploying from a Nexus Docker Registry or GE's DTR on a deployment server, Update the YAML file path by updating the repo url details.
5. Configure the deployment settings, such as paths to the Docker registry, container names, and components to be deployed.
6. Configure host and port values for the OMS Interface Server.
7. Configure the following Enterprise OMS application and database settings:
 - In the `persistentVolume` section, set the actual Local Persistent Volume storage name in the `storageClass` property.
 - In the `database_properties` section, specify the database name, user, and password for database user.

After running a Docker container with the database, you will create a database with these credentials.

- In the service section, specify the port for external database connection.
- In the controller section, specify whether the HTTPS connection with TLS termination is enabled.

Note

A user with the specified credentials must be created in the PostgreSQL database after deploying Enterprise OMS.

1. Specify whether the secure communication is enabled in the "tls" section.
2. Make other changes, as appropriate.

3.1.8. Helm Based Deployment

Example of an YAML File for Custom Namespace

```
namespaces:
- name: foundation-cluster-monitoring
  istio_injected: true
- name: foundation-cluster-operators
  istio_injected: true
- name: foundation-cluster-zerotrust
  istio_injected: true
- name: kyverno
  istio_injected: false
- name: foundation-cluster-tools
  istio_injected: true
- name: foundation-env-default
  istio_injected: true
## add additional namespaces below - the above namespaces are included in the default values file
and must be present - example below to create conjur namespace with sidecar injection enabled
- name: work-model
  istio_injected: true
```

- add helm repository

The example command below assumes digital grid artifactory is used as the helm repository, and that the corresponding user and password are stored in respective environment variables `${GRIDART_USR}` and `${GRIDART_PSW}`

In airgap environments, replace <https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm> with the url to your local helm repository. And pass in the local helm repo's username and password accordingly.

```
helm repo add adms-helm https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm
--username ${GRIDART_USR} --password ${GRIDART_PSW}
```

```
./kubectl create secret docker-registry ${GRIDART_SECRET_NAME} --docker-server=${DGA_REGISTRY}
--docker-username=${GRID_ARTIFACTORY_CREDS_USR} --docker-password=${GRID_ARTIFACTORY_CREDS_PSW} -n
"${OA_DEPLOYMENT_NAMESPACE}" || true
./kubectl patch serviceaccount default -n "${OA_DEPLOYMENT_NAMESPACE}" -p '{"imagePullSecrets":
[{"name": "docker-registry"}]}'
# create PG secret
./kubectl get secret postgres-db-pg-cluster-postgres-secret -n "${PLATFORM_NAMESPACE}" -o yaml |
grep -v '^\\s*namespace:\\s' | kubectl apply -n "${OA_DEPLOYMENT_NAMESPACE}" -f - || true
```

Change to directory where YAMI file is present

```
export VALUES_ARGS="--values work-model-values.yaml --set
global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS},--set global.zitadel.enabled=true,ois-
external-gateway.oisExternalGatewayRoute.endpoints=http://${WIN_SERVER}:8802,oisuaa-external-
gateway.oisuaaExternalGatewayRoute.endpoints=https://${WIN_SERVER}:8812"
helm upgrade -i work-model adms-helm/work-model -n work-model --version 16.18.0 --wait --timeout=15m
\\$VALUES_ARGS
```

3.1.9. ArgoCD Based Deployment

Note

Unlike the helm-only use case, ArgoCD deployment requires pushing the configuration manifests to a Git repository additionally. In either case, make a note of the location of each configuration manifest in Git, as they will be used to configure the work-model ArgoCD application.

The command below assumes that your kubectl client is configured with the argocd cluster kubeconfig file, meaning you have access to it. If not, please contact the argocd admin user/team to obtain the argocd credentials and log in using the classical way.

```
kubectl config set-context --current --namespace=argocd
```

```
argocd login --core
```

- add helm repository

The example command below assumes digital grid artifactory is used as the helm repository, and that the corresponding user and password are stored in respective environment variables \${GRIDART_USR} and \${GRIDART_PSW}

In airgap environments, replace <https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm> with the url to your local helm repository. And pass in the local helm repo's username and password accordingly.

```
helm repo add adms-helm https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm
--username ${GRIDART_USR} --password ${GRIDART_PSW}
./kubectl create secret docker-registry ${GRIDART_SECRET_NAME} --docker-server=${DGA_REGISTRY}
```

```

--docker-username=${GRID_ARTIFACTORY_CREDS_USR} --docker-password=${GRID_ARTIFACTORY_CREDS_PSW} -n
"${OA_DEPLOYMENT_NAMESPACE}" || true
./kubectl patch serviceaccount default -n "${OA_DEPLOYMENT_NAMESPACE}" -p '{"imagePullSecrets":
[{"name": "docker-registry"}]}'
# create PG secret
./kubectl get secret postgres-db-pg-cluster-postgres-secret -n "${PLATFORM_NAMESPACE}" -o yaml |
grep -v '^\\s*namespace:\\s' | kubectl apply -n "${OA_DEPLOYMENT_NAMESPACE}" -f - || true
argocd repo add https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/adms-helm \
  --type helm \
  --name adms-helm \
  --username ${GRIDART_USR} \
  --password ${GRIDART_PSW} \
  --upsert

argocd repo add https://github.build.ge.com/IDMS/work-model-charts.git \
  --username ${GIT_USR} \
  --password ${GIT_PSW} \
  --upsert

argocd app create -f work-model.yaml --repo https://github.build.ge.com/IDMS/work-model-charts.git \
  --helm-set-string
spec.sources.values.adhocGlobals.global.clusterExternalUrl=https://${CLUSTER_EXTERNAL_DNS} \
  --helm-set-string spec.sources.values.adhocGlobals.global.zitadel.enabled=true \
  --helm-set-string spec.sources.values.adhocGlobals.ois-external-
gateway.oisExternalGatewayRoute.endpoints=http://${WIN_SERVER}:8802,
  --helm-set-string spec.sources.values.adhocGlobals.oisuaa-external-
gateway.oisuaaExternalGatewayRoute.endpoints=https://${WIN_SERVER}:8812"
  --upsert
argocd app wait work-model --sync --timeout 600
argocd app wait work-model --operation --timeout 600
argocd app wait work-model --health --timeout 600

```

Example of an YAML File for Deploying Foundation Base

```

spec:
  destination:
    server: "https://kubernetes.default.svc"
  sources:
    helmRepo: "https://dig-grid-artifactory.apps.ge.com/artifactory/api/helm/foundation-helm"
    values:
      repo: "https://github.build.ge.com/grid-foundation/foundation-reference-app"
      targetRevision: "2024-03"
      foundationProfile:
        prefixPath: "release/foundation-helm-profiles"
        name: "default"

#### sync-wave: 10
kyvernoRancherCrds:
  chartName: "kyverno-rancher-crds"
  #overrideTargetRevision: 1.2.0

```

```

#### sync-wave: 20
foundationPoliciesCerts:
  chartName: "foundation-policies-certs"
  #overrideTargetRevision: 1.2.0

#### sync-wave: 30
zeroTrustOperator:
  chartName: "zero-trust-operator"
  #overrideTargetRevision: 1.2.0

#### sync-wave: 40
istioPrep:
  chartName: "istio-prep"
  #overrideTargetRevision: 1.2.0

#### sync-wave: 40
istiod:
  chartName: "istiod"
  #overrideTargetRevision: 1.2.0

#### sync-wave: 50
foundationOperators:
  chartName: "foundation-operators"
  #overrideTargetRevision: 1.2.0

#### sync-wave: 60
monitoringApps:
  chartName: "monitoring-apps"
  #overrideTargetRevision: 1.2.0

#### sync-wave: 70
k8sIngressNginx:
  chartName: "k8s-ingress-nginx"
  #overrideTargetRevision: 1.2.0

#### sync-wave: 80
zeroTrustApps:
  chartName: "zerotrust-apps"
  #overrideTargetRevision: 1.2.0

#### sync-wave: 81
foundationAuthWS:
  chartName: "foundation-auth-ws"
  #overrideTargetRevision: 1.0.0

#### sync-wave: 90
velero:
  chartName: "velero"
  #overrideTargetRevision: 1.2.0

#### sync-wave: 100
foundationClusterTools:
  chartName: "foundation-cluster-tools"

```

```
#overrideTargetRevision: 1.2.0
```

Example of an YAML File for Deploying Enterprise OMS

YAML file of work-model.yaml

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: work-model
  namespace: argocd
  annotations:
    argocd.argoproj.io/sync-wave: "1"
  finalizers:
    - resources-finalizer.argocd.argoproj.io
spec:
  destination:
    namespace: work-model
    server: https://kubernetes.default.svc
  project: default
  source:
    helm:
      # parameters:
      # - name: global.registry
      #   value: dig-grid-artifactory.apps.ge.com
      valueFiles:
        - ../../apps-config/work-model-values.yaml
    path: charts/work-model-all
    repoURL: https://github.software.governova.com/grid-operate/work-model-charts.git
    targetRevision: FRA-VERSION
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
    syncOptions:
      - CreateNamespace=true
      - RespectIgnoreDifferences=true
```

YAML file of work-model-values.yaml

```
# This is a YAML-formatted file.
# Declare variables to be passed into your templates.

# This is a YAML-formatted file.
# Declare variables to be passed into your templates.

global:
  workModelNamespace: work-model
  gatewayroute: true # Gatewayroute service to support foundation base deployment
  zeroTrustNamespace: foundation-cluster-zerotrust
  fqdnSuffix: .svc.cluster.local
  jwksUri: http://sws-uaa.foundation-cluster-zerotrust.svc.cluster.local:8080/uaa/token_keys
```

```

clusterExternalUrl: http://ingress.local
zitadel:
  enabled: true
zitadelProxyDns: ""
serviceProxyDns: ""
TZ: "UTC"
image:
  registry: "dig-grid-artifactory.apps.ge.com/virtual-docker"
  pullPolicy: Always
imagePullSecrets:
  - name: docker-registry

```

#Postgresql configuration

```

postgres:
  serviceHost: postgres-db-pg-cluster.foundation-cluster-zerotrust.svc.cluster.local
  servicePort: 5432
  wmCreateSecret: true
  wmDbSecret: pg-work-model-secret
  workModeldb:
    username: wmpostgres
    password: admin1
    upgradedatabase: 'false'
  pgSecretName: postgres-db-pg-cluster-postgres-secret
  securityContext:
    runAsNonRoot: true
    runAsUser: 1337

```

ois-external-gateway: # Configure multiple OMS Interface Servers with UAA-disabled (required for eOMS)

```

enabled: true
oisExternalGatewayRoute:
  endpoints:
    - http://${WIN_SERVER}:8802 #OIS Server 1 (uaa-disabled)
    # - https://172.227.51.85:8802 #OIS Server 2 (uaa-disabled)
    # - https://172.227.51.100:8802 #OIS Server 3 (uaa-disabled)
    # - https://172.227.51.150:8802 #OIS Server 4 (uaa-disabled)

```

admsExternalGateway:

```

  loggingLevel: ERROR
  replicas: 1

```

admsCertSecret: # Certificate to connect to the OMS Interface Server (ADMS box)

```

cert: |
  -----BEGIN CERTIFICATE-----
  MIIDozCCAougAwIBAgIUU3KvyEumb+TRqvDf5D9kxCipiKwwDQYJKoZIhvcNAQEL
  BQAwYTELMAkGA1UEBhMCQVUxEzARBgNVBAgMC1NvbWUtU3RhdGUxEzARBgNVBAoM
  CkdFIERpZ210YWwKDAkGA1UzBAMH0RFRMi4wIEZvdW5kYXRpb24gQmFzZSAIFJv
  b3QgQ0EwHhcNMjEwMDY0OTQ0WhcNMjEwMDY0OTQ0WjBhMQswCQYDVQQGEwJBTETMBE
  GA1UECwKU29tZS1TdGF0ZTETMBEGA1UECgwKR0UgRGlnaXRhbDEoMCYGA1UEAwwFREUy
  LjAgRm91bmRhdG1vbiBCYXNlIC0gUm9vdCBBDQTCASIAwDQYJKoZIhvcNAQEBBQADggEP
  ADCCAQoCggEBAOrcZK4NB0QuunFJ8Di5bhl+PDwP6fDdZdx06w0t0vguq2KagDFJjHa0NyChLHLJQdKHQ+ga2kY4Wqxt1w0lhCvC0gxVYQwV
  IHVinzbFD1/KxUGy6n2AIXhcGTLl+pbYL6e54ls1+T+orN76YYCt72o6cZlpwLh
  ODfTfLX4sP/ljLYJmIw/X0dR85NoDiAVtm2S9pcdl0/ivorJnlq39LJ+LgJgIkG/
  FRJpMiL/yqOTwnizq/M9kaM8HCLNUYRzM+m8eIIXCM7hJrp060GWL3ken6BW0QXf
  NPPw9CWpRxTHAZcWsInvoCY+MwD3g9mQ6xHvzbXjgYIoVNMwEqHydkCAwEAaANT

```

```

MFEwHQYDVR00BBYEFAaNoDyDSLFFbNxaBJ7PDKFW0rQ0MB8GA1UdIwQYMBaAFAaN
oDyDSLFFbNxaBJ7PDKFW0rQ0MA8GA1UdEwEB/wQFMAMBAf8wDQYJKoZIhvcNAQEL
BQADggEBAEMA0F5u3aiHyfNDJDV3L2me2e1eKTqojr7oMHDNGj0JNqtJ7A1PMwor
RutqYY6Jwe3j08fuIyF9EezS0305LgLS683jkN3KidwSX1NpDE4aMo1X1TxA+12
LwbXXLTORMOKP1FQH1kRgaTKQKnqCbW1Jet/AyR9DIiPjSjPwVJJm3Pm1ztg4w0e
jfkDhMU1QiqSQ7aviaC0Kiy8VYiDQLo0GitrFo+C+uaQKXwCAhRIRBZbLnBgFP6Q
SWtS9o4M0f1tiqCSUEhT8wTYf2pEqPDMtEx5res8sdg+P45cIphFpB7pGojIDzKs
uVZAiK58024KW49o74I+QrMbWqeH0mc=
-----END CERTIFICATE-----

```

This certificate is passed to the cluser via values.yaml file.

oisuaa-external-gateway: # Configure multiple OMS Interface Servers with UAA-enabled (required to work with Outage Assist)

enabled: true

oisuaaExternalGatewayRoute:

endpoints:

- https://\${WIN_SERVER}:8812 #OIS Server 1 (uaa-enabled)

- https://172.227.51.25:8812 #OIS Server 2 (uaa-enabled)

- https://172.227.51.55:8812 #OIS Server 3 (uaa-enabled)

- https://172.227.51.200:8812 #OIS Server 4 (uaa-enabled)

enabled: true

admsExternalGateway:

loggingLevel: ERROR

replicas: 1

admsCertSecret: # Certificate to connect to the OMS Interface Server (ADMS box)

cert: |

-----BEGIN CERTIFICATE-----

```

MIIDoZCCAougAwIBAgIUU3KvyEumb+TRqvDf5D9kxCipiKwwDQYJKoZIhvcNAQEL
BQAwYTELMakGA1UEBhMCVUxEzARBgNVBAgMClNvbWUtU3RhdGUxEzARBgNVBAoM
CkdFIERpZ210YVwwKDAmbG9NVBAMMH0RFMi4wIEZvdW5kYXRpb24gQmFzZSAIFJv
b3QgQ0EwHhcNMjEwMDY0OTQ0WhcNMzEwMDY0OTQ0WjBhMQswCQYDVQQG
EwJBVTETMBEGA1UECAwKU29tZS1TdGF0ZTETMBEGA1UECgwKR0UgRGlNaXRhbDEo
MCYGA1UEAwwFREUyLjAgRm91bmRhG1vbiBCYXN1IC0gUm9vdCBBDQTCCASwDQYJ
KoZIhvcNAQEBBQADggEPADCCAQoCggEBA0rcZK4NB0QuunFJ8Di5bhl+PDwP6fDd
Zdx06w0tOvguq2KagDFJjHa0NyChLHLJQdKHQ+ga2kY4Wqxt1w0lhCvC0gxVYQwV
IHVinzbFD1/KxUGy6n2AIxhcGTL+pbYL6e54lsl+T+orN76YYCt72o6cZlpwLh
ODfTfLX4sP/ljLYJmIw/X0dR85NoDiAVtm2S9pcdl0/ivorJnlq39LJ+LgJgIkG/
FRJpMil/yqOTwnizq/M9kaM8HCLNUYRzM+m8eIIXCM7hJrp060GWL3ken6BW0QXf
NPPw9CWpRxTHAZcWsInvoCY+MwD3g9mQ6xHvzbXjgYIoVNMwEqHydkCAwEAANT
MFEwHQYDVR00BBYEFAaNoDyDSLFFbNxaBJ7PDKFW0rQ0MB8GA1UdIwQYMBaAFAaN
oDyDSLFFbNxaBJ7PDKFW0rQ0MA8GA1UdEwEB/wQFMAMBAf8wDQYJKoZIhvcNAQEL
BQADggEBAEMA0F5u3aiHyfNDJDV3L2me2e1eKTqojr7oMHDNGj0JNqtJ7A1PMwor
RutqYY6Jwe3j08fuIyF9EezS0305LgLS683jkN3KidwSX1NpDE4aMo1X1TxA+12
LwbXXLTORMOKP1FQH1kRgaTKQKnqCbW1Jet/AyR9DIiPjSjPwVJJm3Pm1ztg4w0e
jfkDhMU1QiqSQ7aviaC0Kiy8VYiDQLo0GitrFo+C+uaQKXwCAhRIRBZbLnBgFP6Q
SWtS9o4M0f1tiqCSUEhT8wTYf2pEqPDMtEx5res8sdg+P45cIphFpB7pGojIDzKs
uVZAiK58024KW49o74I+QrMbWqeH0mc=
-----END CERTIFICATE-----

```

This certificate is passed to the cluser via values.yaml file.

uaaClient:

oisUaaClient:

authorities: # Needs to match etd group defined in roles-config YAML


```

- service.account.group5

work-model-eventhandler:
  replicaCount: 1
  enabled: true
  service:
    name: work-model-eventhandler
    port: 8080
    port_name: event-port
  env:
    EUREKA_URI: "http://eureka-1:8761/eureka"
    EUREKA_CLIENT_ENABLED: "false"
    SERVER_PORT: "8080"
    SERVICES_DMSINTERFACESERVER_URL: "http://dmsinterfaceserver:8801"
    SERVICES_OMSINTERFACESERVER_URL: "http://ois-external-gateway/ois"
    SERVICES_OMSINTERFACESERVER_UAAURL: "http://oisuaa-external-gateway/oisuaa"
    SERVICES_WORKMODELSERVICE_URL: "http://work-model-service:8080"
    SPRING_PROFILES_ACTIVE: "kubernetes,debug"
    FAULTLOCATION_CREATETASK: "true"
    ORDERS_AUTOCONFIG_MANUALSWITCHING: 2
    ORDERS_AUTOCONFIG_PLANNEDSWITCHING: 2
    ORDERS_AUTOCONFIG_LOADSHED: 2
    DISABLE_FIRST_ORDER_POST_FIX_ID: "false"

work-model-notifications:
  # This is a YAML-formatted file.
  # Declare variables to be passed into your templates.
  enabled: true
  replicaCount: 1
  service:
    name: work-model-notifications
    port: 8080
    port_name: notify-port
  env:
    EUREKA_URI: "http://eureka-1:8761/eureka"
    HAZELCAST_CLIENT_ADDRESSES: "work-model-service:5701"
    EUREKA_CLIENT_ENABLED: "false"
    SERVER_PORT: "8080"
    SPRING_PROFILES_ACTIVE: "kubernetes,debug"

work-model-service:
  enabled: true
  replicaCount: 1
  service:
    name: work-model-service
    port: 8080
    port_name: service-port
  hazelcast:
    port: 5701
    port_name: hazelcast

```

```

env:
  SPRING_DATASOURCE_URL: jdbc:postgresql://postgres-db-pg-cluster.foundation-cluster-
zerotrust.svc.cluster.local:5432/workmodel?useSSL=false
  EUREKA_URI: "http://eureka-1:8761/eureka"
  EUREKA_CLIENT_ENABLED: "false"
  SERVER_PORT: "8080"
  SPRING_PROFILES_ACTIVE: "kubernetes,debug"
  K8S_SERVICE_NAME: work-model-service
  K8S_SERVICE_NAMESPACE: default
  K8S_SERVICE_HAZELCASTPORT: 5701
  EVENTS_CLEAN_AGE: "PT1H"
  EVENTS_CLEAN_DELAY: "PT10M"
  REMOVEORDERS_REMOVE_INITIAL_DELAY: "PT1M"
  REMOVEORDERS_REMOVE_DELAY: "PT24H"
  REMOVEORDERS_REMOVE_RETENTION_PERIOD: "PT1080H"

```

3.1.10. Setting Up Firewall Rules

Open the firewall ports listed in table below to allow communication among the Enterprise OMS Kubernetes cluster and ADMS servers.

Table 1. Firewall Ports to Open When Installing in Kubernetes

Source	Destination	Port	Protocol
Work Order Adapter (on the ADMS Interface Server host)	Work Model Event Handler Service (in the Enterprise OMS Kubernetes cluster)	8080	HTTPS/WSS
NetServer, APF, Browser (on the ADMS Server host)			

i Note

In the Kubernetes cluster, all application traffic is redirected internally via the envoy proxy. Therefore, you must only have port 8080 open.

3.2. Preparing for the Installing in Windows

Generate certificates as described in [Generating Certificates](#).

3.2.1. Generating Certificates

Secure connections between the Enterprise OMS and the ADMS system is established using an HTTPS connection with TLS termination.

For HTTPS with TLS, you need to obtain SSL certificates from your certificate authority. The certification authority is based on Active Directory or other services. Therefore, you need to have two certificates: one for Kubernetes Ingress Controller and another for the ADMS interface server.

For Enterprise OMS, you need the following certification files:

- Root certificate authority certificate: This certificate is used to issue and sign certificates. (For example, file name = ca.p12; CN = Work Model Root CA).
- Trust Store file (cacerts): The cacerts file is delivered with the Java Runtime Environment and the Java Development Kit. It is typically located at JAVA_HOME/lib/security/cacerts. That file contains all the standard Trusted Certificate Authorities for the internet.

You can copy cacerts, and then add the Certificate Authority that is used to sign all the Enterprise OMS certificates. If a trusted Certificate Authority is used, then only a copy of the cacerts is required. If a Self-Signed Certificate Authority is created, then that Self-Signed Certificate Authority must be added to the cacerts file that was copied.

- ADMS server certificate: This certificate is used for secure communication between the Kubernetes cluster and ADMS interface servers. (For example, file name = keystore.p12; CN = IDMS)
- PostgreSQL certificate: This certificate is used for establishing secure connections with the PostgreSQL database. (For example, file name = postgres.pem; CN=postgres)

If SSL is enabled for PostgreSQL, you need the root certificate authority certificate (ca.p12) and ADMS server certificate and key (keystore.p12) on the ADMS Interface Server host. You also need the PostgreSQL certificate (postgres.p12) on the Enterprise OMS host to enable the Work Model Event Handler service to connect to the PostgreSQL server.

For more information about configuring PostgreSQL SSL, see [Configuring Database SSL Configuration](#).

For production, it is recommended to use certificates issued by a certificate authority. For testing and demo purposes, you can use self-signed certificates.

The example below shows how to generate OpenSSL certificates using the genrsa tool. This is a standard tool for most Linux distributions. For more information, see <https://www.openssl.org/source/>. The PostgreSQL certificate is generated using the "keytool" application, which is part of the Java Development Kit (JDK) 17.

For more information about generating and using certificates, see [Generating an SSL/TLS Certificate](#).

To generate self-signed certificates:

1. Generate a root certificate authority certificate and key:

```
openssl genrsa -out rootCA.key 2048
openssl req -x509 -new -nodes -key rootCA.key -sha256 -days 3650 -out rootCA.crt -subj
"/C=US/ST=FL/O=GE Digital/CN=Work Model CA"
keytool -import -trustcacerts -keystore "%JAVA_HOME%\lib\security\cacerts" -storepass changeit
-alias WorkModelCA -file rootCA.crt
```

2. Add the Self-Signed Certificate Authority to the cacerts file:

```
copy %JAVA_HOME\lib\security\cacerts .
keytool -import -trustcacerts -keystore .\cacerts -storepass changeit -alias WorkModelCA -file
rootCA.crt
```

3. Generate a certificate and key for the Enterprise OMS services:

Note

It is recommended to set CN to the actual ADMS DNS host name (web address you access when connecting to a secure site).

```
openssl genrsa -out service.key 2048
openssl req -new -sha256 -key service.key -subj "/C=US/ST=CA/O=GE Digital/CN=localhost" -out
service.csr
openssl x509 -req -in service.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out
service.crt -days 3650 -sha256
openssl pkcs12 -export -nodes -out keystore.p12 -inkey service.key -in service.crt -certfile
rootCA.crt -name service -passout pass:
```

4. Generate a certificate for PostgreSQL:

The certificate generated for connecting to the PostgreSQL database is like the certificated generated for the other Enterprise OMS services with one exception. The Common Name (CN) for the Certificate must be the same as POSTGRES_USER. For example, if the username to connect to the database is called "postgres", the CN value for the certificate must be CN=postgres.

```
keytool -genkeypair -alias postgres -dname "CN=postgres" -validity 3650 -keyalg RSA -keysize 2048
-keystore postgres.p12 -storetype PKCS12 -keypass password -storepass password
keytool -keystore postgres.p12 -storepass password -certreq -alias postgres | keytool -keystore
ca.p12 -storepass password -gencert -alias ca -ext ku:c=dig,keyEnc -ext eku=sa,ca -rfc >
postgres.pem
keytool -keystore postgres.p12 -noprompt -storepass password -importcert -trustcacerts -alias ca
-file ca.pem
keytool -keystore postgres.p12 -noprompt -storepass password -importcert -alias postgres -file
postgres.pem
# Install the client keypair into the work-model-service
mv postgres.p12 ./work-model-service/postgres.p12
```

5. Export the server.crt, postgres.crt certificates, and the "server.key" key from the PKCS12 keystore containers:

```
keytool -exportcert -alias service -keystore postgres.p12 -storepass password -rfc -file
postgres.crt
keytool -exportcert -alias service -keystore keystore.p12 -storepass password -rfc -file
server.crt
openssl pkcs12 -in keystore.p12 -nodes -nocerts -out server.key
```

6. When installing in Windows and using a Self-Signed Certificate Authority, copy the Trust Store

file (cacerts) and the ADMS server certificate to a local folder.

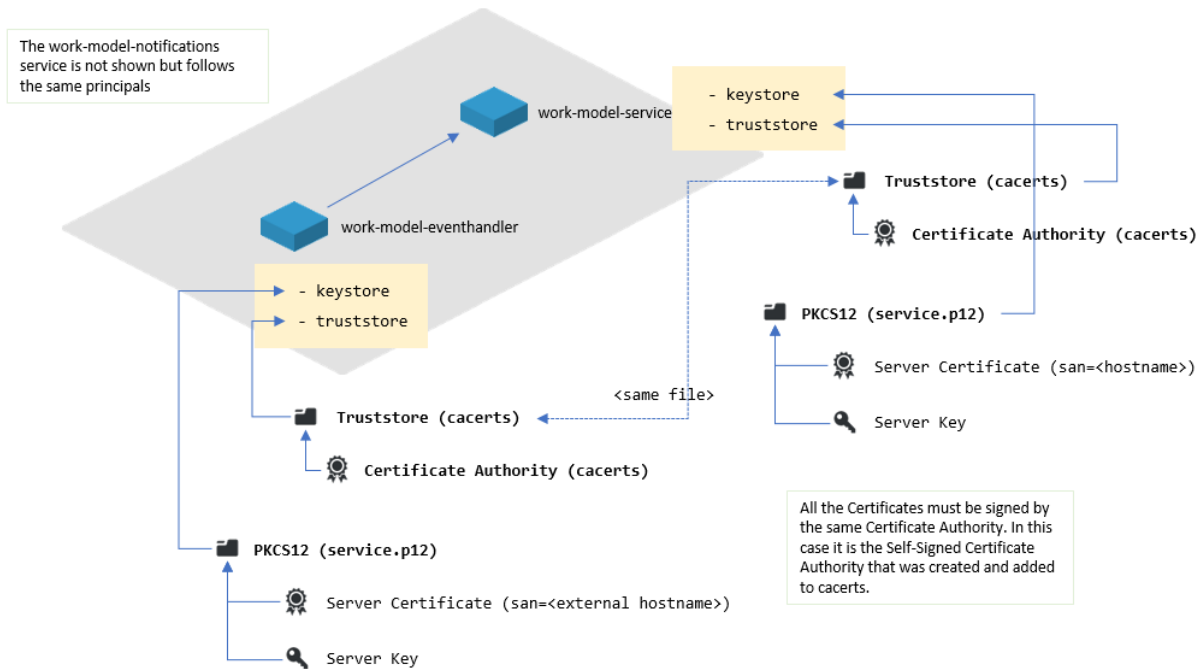


Figure 6. Certificate Flow for Enterprise OMS Services

3.2.2. Setting Up Firewall Rules

Open the firewall ports listed in previous table to allow communication among the Enterprise OMS server and ADMS servers.

Table 2. Firewall Ports to Open When Installing in Windows

Source	Destination	Port	Protocol
PostgreSQL DB (on the Enterprise OMS server host)	ADMS Interface Server	5432	TCP
Work Model Event Handler Service (on the Enterprise OMS server host)	OMS Interface Server (on the ADMS Interface Server host)	8802	HTTPS
Work Order Database Sync (on the ADMS Interface Server host)	Work Model Notifications Service (on the Enterprise OMS server host)	8083	HTTPS/WSS
NetServer (on the ADMS Server host)			
Work Order Adapter (in ADMS Interface Server)	PostgreSQL DB (on the Enterprise OMS server host)	30001	HTTPS/WSS
Work Order Adapter (on the ADMS Interface Server host)	Work Model Event Handler Service (on the Enterprise OMS server host)	8080	HTTPS/WSS
NetServer, APF, Browser (on the ADMS Server host)			

Note

Port numbers can be configured in configuration files for the associated applications.

3.3. Configuring EOMS Components for a Multi-Host System

For more information about configuring ADMS components for a multi-host system setup, see [Configuring ADMS Components for a Multi-Host System](#).

3.3.1. Windows Configuration

For Windows configuration, do the following:

1. Edit the work-model-eventhandler.properties file (by default, stored in Distribution\WorkModelServices\config\):

```
server.port=8080

dataservice=https://localhost:9090

ois=https://ADMS1:8802;http://ADMS2:8802

oisuaa=http://localhost:8812

ssl.enabled=true

keystore=${WM_HOME}\\keystore.p12
keystore.alias=service
truststore=${WM_HOME}\\cacerts

# -----
# Order Auto Configuration
#   Uncomment to override the order auto close configuration
#   # possible values: 0 = do nothing, 1 auto field complete, 2 auto close
#   # default : 2 (auto close)
# -----
# ORDERS_AUTOCONFIG_MANUALSWITCHING=2
# ORDERS_AUTOCONFIG_PLANNEDSWITCHING=2
# ORDERS_AUTOCONFIG_LOADSHED=2
...
```

2. Edit the work-model-notifications.properties file (by default, stored in Distribution\WorkModelServices\config\):

```
server.port=8083
```

```
hazelcast=localhost:5701

ssl.enabled=true

keystore=${WM_HOME}\\keystore.p12
keystore.alias=service
truststore=${WM_HOME}\\cacerts
```

3. Edit the work-model-service.properties file (by default, stored in Distribution\WorkModelServices\config\):

```
server.port=9090

database.host=localhost
database.port=5432
database.name=workModel
database.username=postgres
# The password can be encrypted by running work-model-service.jar with the argument encrypt-
password
# Example "java -jar work-model-service.jar encrypt-password"
# Entering the Password will result in an encrypted password; for example,
ENC(0m/1LP6f4DymtrxDtLX6eoALY0+In/du7UBfUv03ofVmp9hM1WTPDw==)
# This encrypted password can be used for the property database.password
database.password=ENC(0m/1LP6f4DymtrxDtLX6eoALY0+In/du7UBfUv03ofVmp9hM1WTPDw==)

# -----
# Database TLS Configuration
# Uncomment if postgres is configured to require SSL
# -----

database.ssl.enabled=true
database.keystore=${WM_HOME}\\postgres.p12
database.keystore.password=
database.truststore=${WM_HOME}\\cacerts
database.truststore.password=

ssl.enabled=true

keystore=${WM_HOME}\\keystore.p12
keystore.alias=service
truststore=${WM_HOME}\\cacerts
...
```

3.3.2. Kubernetes Configuration

Prepare dual Kubernetes clusters with EOMS.



Note

Work Model services (EOMS) are not site-aware. In a Kubernetes environment, run these services only on the enabled or active node. If the enabled or active node switches to standby, the services stop on that node and start on the newly enabled node.

4. Configuring ADMS

4.1. Configuring the hosts.etc File

Important

Perform the following operation for each machine (host) where ADMS interface servers (OMS Interface Server and Work Order Adapter) are running.

- In the hosts.etc file, add the following entry:

```
work-model <IP address of the Enterprise OMS host>
```

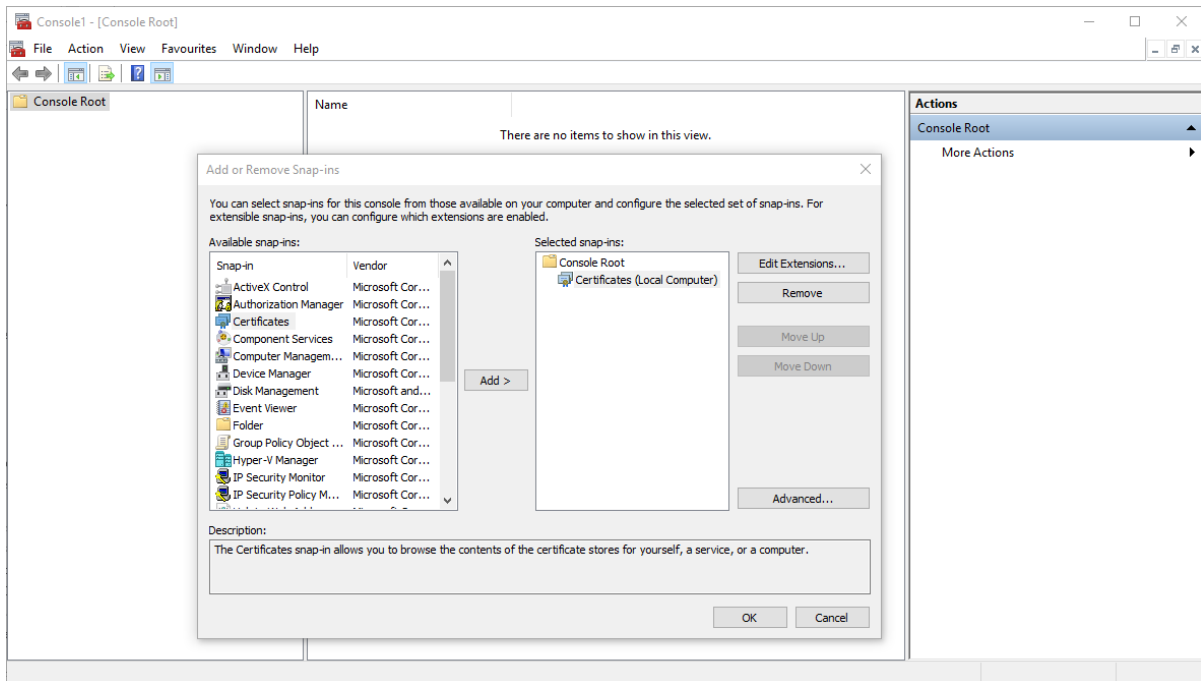
4.2. Importing Certificates to the ADMS Hosts

Important

Perform the following steps for each machine (host) where the ADMS interface servers (OMS Interface Server and Work Order Adapter) are running.

To import certificates to an ADMS interface server host:

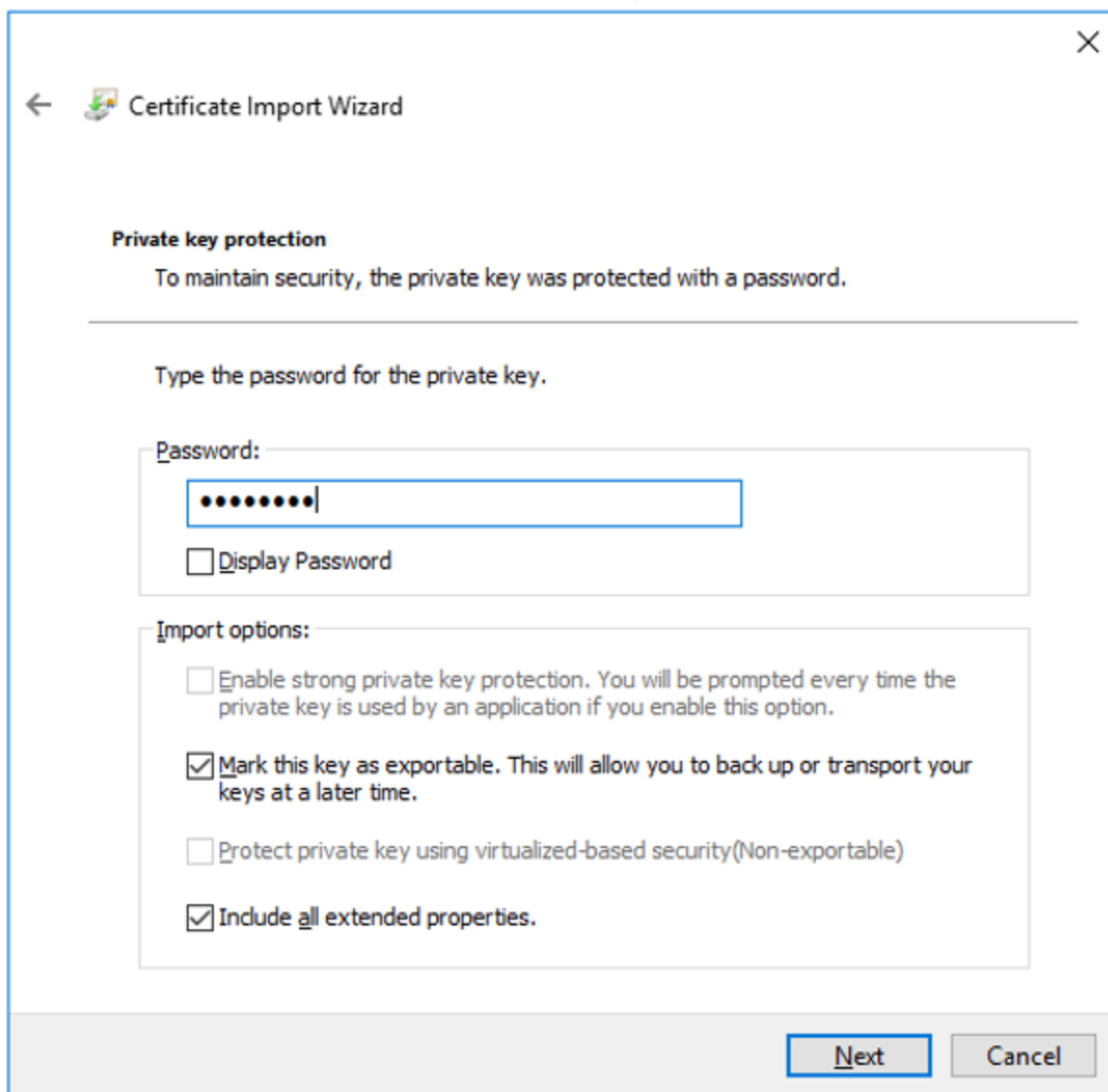
1. Generate certificates as described in [Generating Certificates](#).
2. Copy the Root certificate authority certificate (ca.p12) and the Enterprise OMS service certificate (keystore.p12) to the ADMS machine.
3. Open the Microsoft Management console.
4. From the File menu, select Add/Remove Snap-in.
5. From the available snap-ins, select Certificates and click Add.
6. Select the Computer Account option and click Next.
7. Select the Local Computer option and click Finish.



8. Click OK.
9. In Console Root, expand the Certificates (Local Computer) list.
10. Right-click Trusted Root Certification Authorities, select All Tasks, and click Import.

The Certificate Import wizard appears.

11. Verify that the Store location is Local Machine and click Next.
12. Click Browse, select the ca.p12 file to import, and click Next.
13. Enter the password that you used when generating the certificate (see [Generating Certificates](#)).
14. Disable the "Strong Private Key Protection" option.
15. Enable the "Mark this key as exportable" option, and click Next..



The image shows a Windows-style dialog box titled "Certificate Import Wizard". It has a back arrow icon on the left and a close "X" icon on the right. The main content area is titled "Private key protection" and contains the text "To maintain security, the private key was protected with a password." Below this is a horizontal line and the instruction "Type the password for the private key." There is a "Password:" label followed by a text input field containing ten dots. Below the input field is a checkbox labeled "Display Password". Below this is another section titled "Import options:" containing four checkboxes: "Enable strong private key protection. You will be prompted every time the private key is used by an application if you enable this option." (unchecked), "Mark this key as exportable. This will allow you to back up or transport your keys at a later time." (checked), "Protect private key using virtualized-based security(Non-exportable)" (unchecked), and "Include all extended properties." (checked). At the bottom right of the dialog are two buttons: "Next" and "Cancel".

← Certificate Import Wizard

Private key protection

To maintain security, the private key was protected with a password.

Type the password for the private key.

Password:

☐ Display Password

Import options:

☐ Enable strong private key protection. You will be prompted every time the private key is used by an application if you enable this option.

☒ Mark this key as exportable. This will allow you to back up or transport your keys at a later time.

☐ Protect private key using virtualized-based security(Non-exportable)

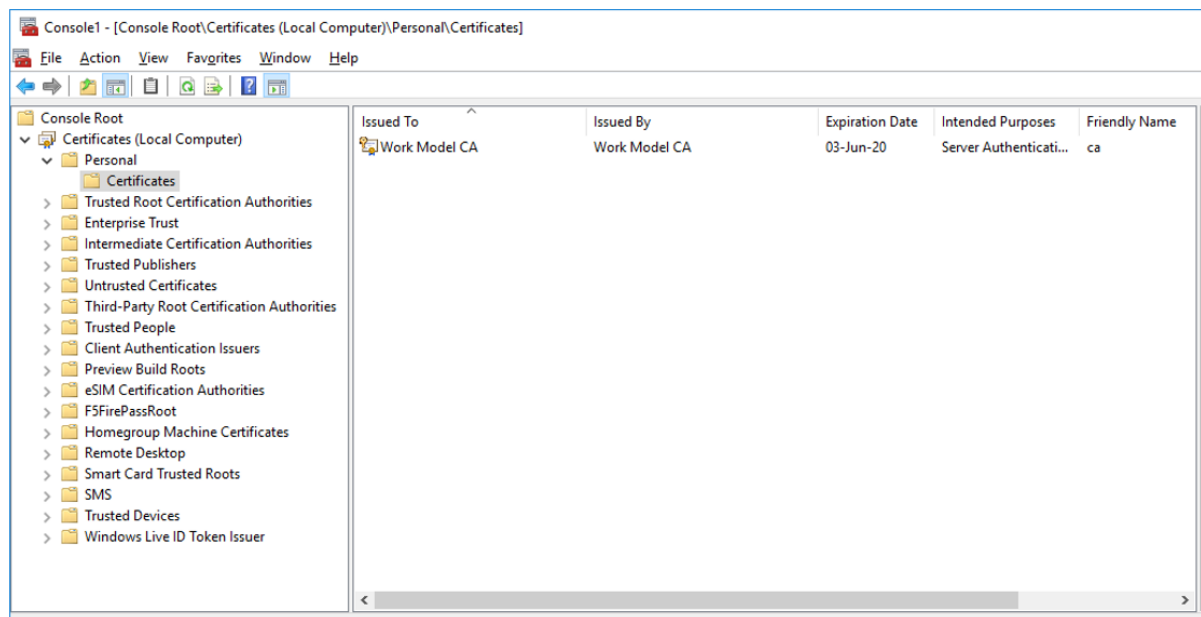
☒ Include all extended properties.

Next Cancel

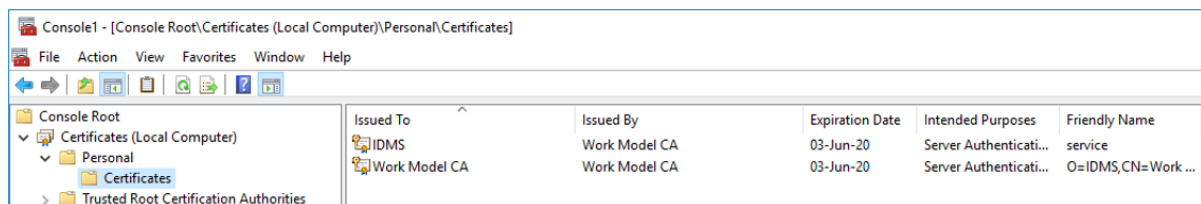
16. Select the Place All Certificates in the Following Store option, select Personal, and click Next.

17. Click Finish.

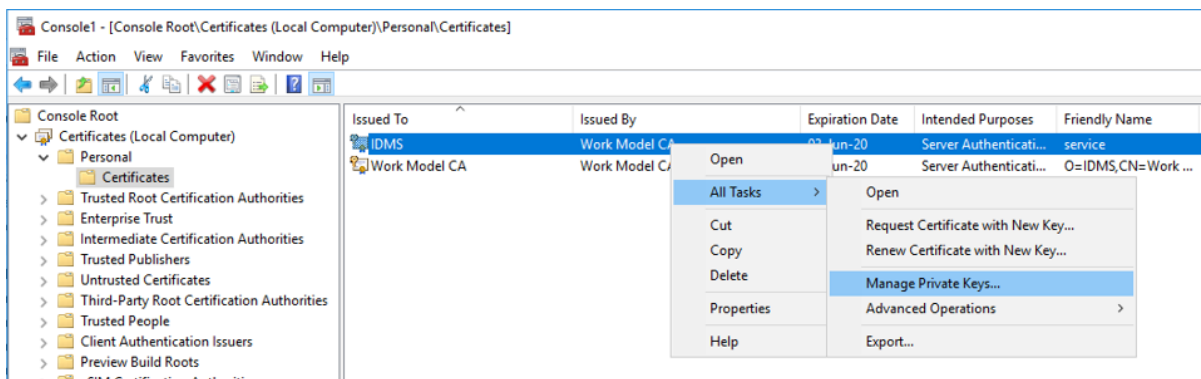
The root certificate appears in the Certificates list.



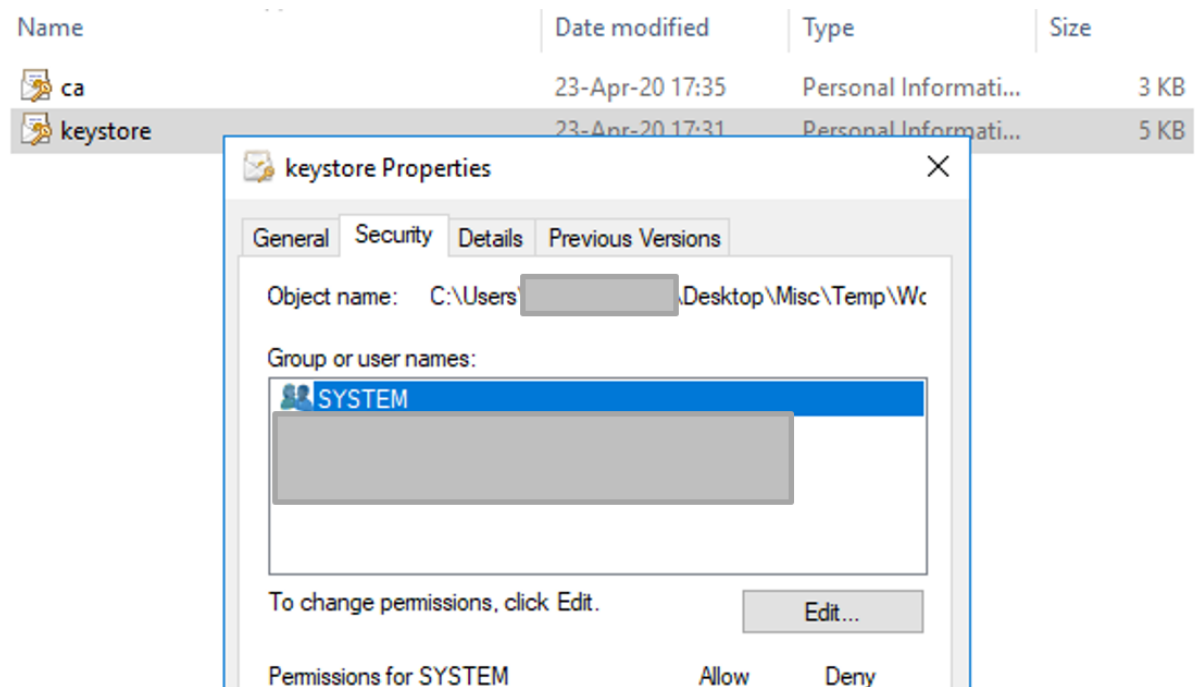
18. Perform the same operations to import the keystore.p12.
19. Verify that the Root certificate authority certificate (Work Model Root CA) and the ADMS server certificate (IDMS) are listed in the Personal folder under Certificates (Local Computer).



20. Right-click the ADMS server certificate (IDMS), point to All Tasks, and click Manage Private Keys.



Alternatively, open the Security tab of the certificate Properties display.

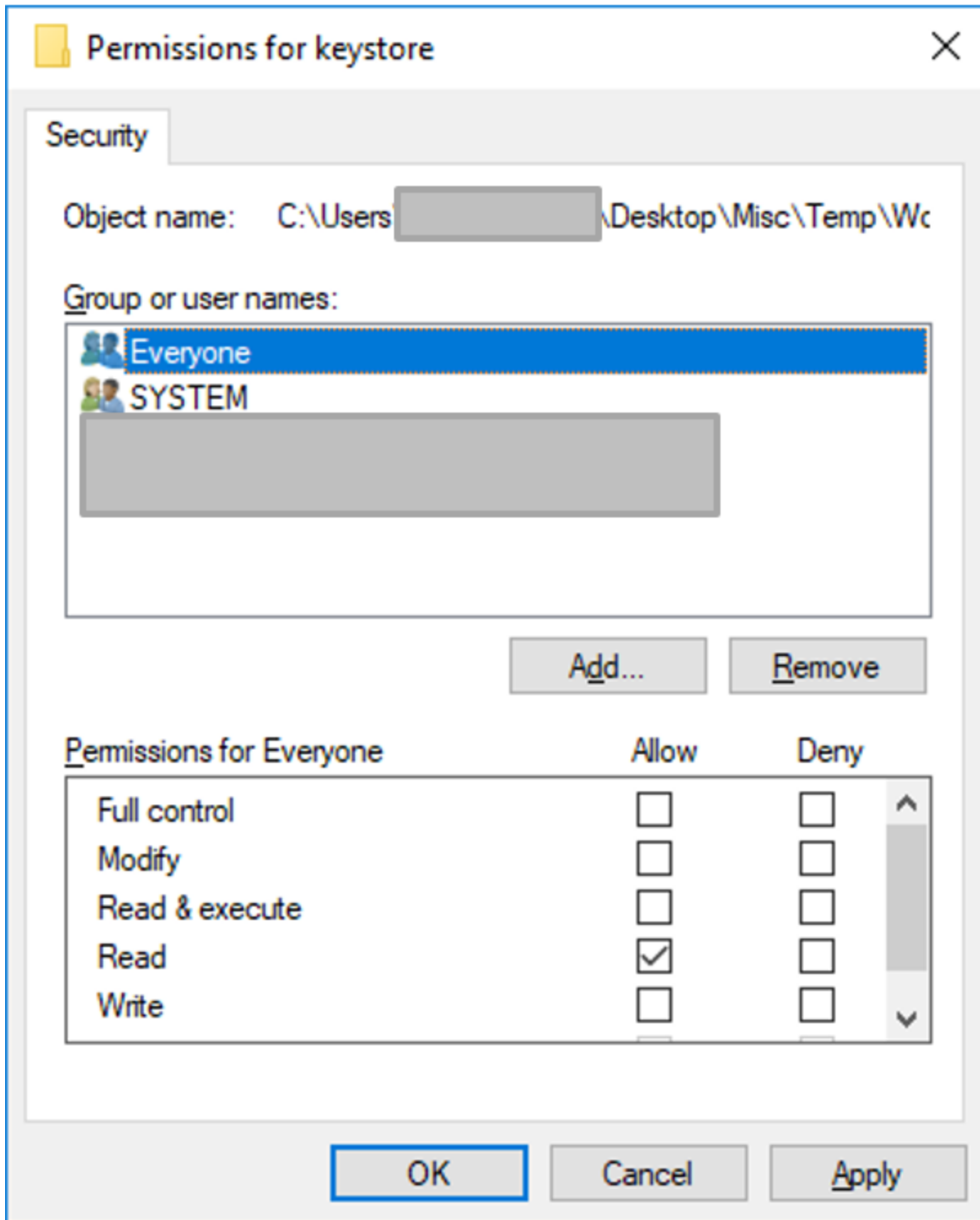


21. Add the user or service account that is used to run ADMS servers (or select Everyone, which is less secure) and give Read permissions.

To read from Trusted Root Certification Authorities, permission must be granted for users that do not have Read permissions.

! Important

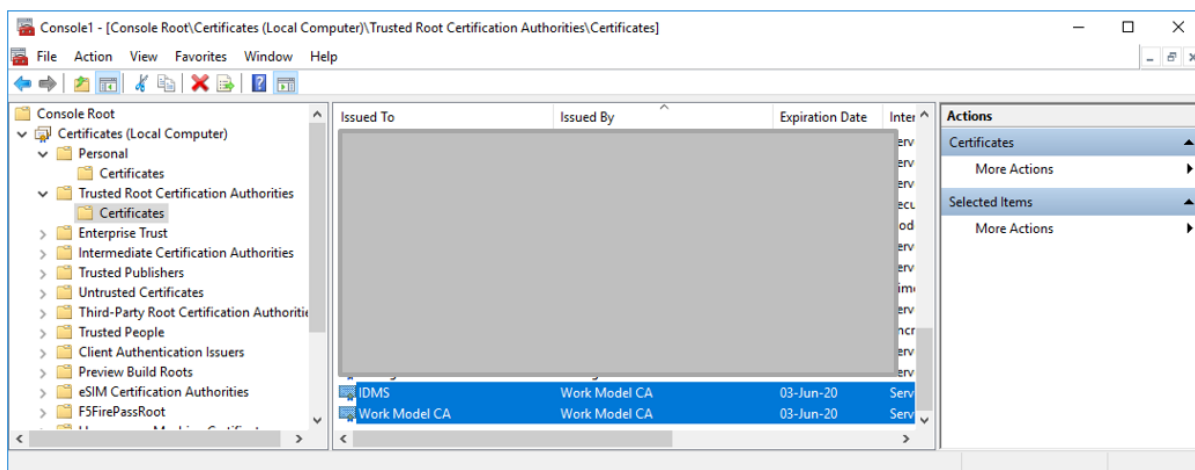
Ensure that you give permissions to the private key, not the public key.



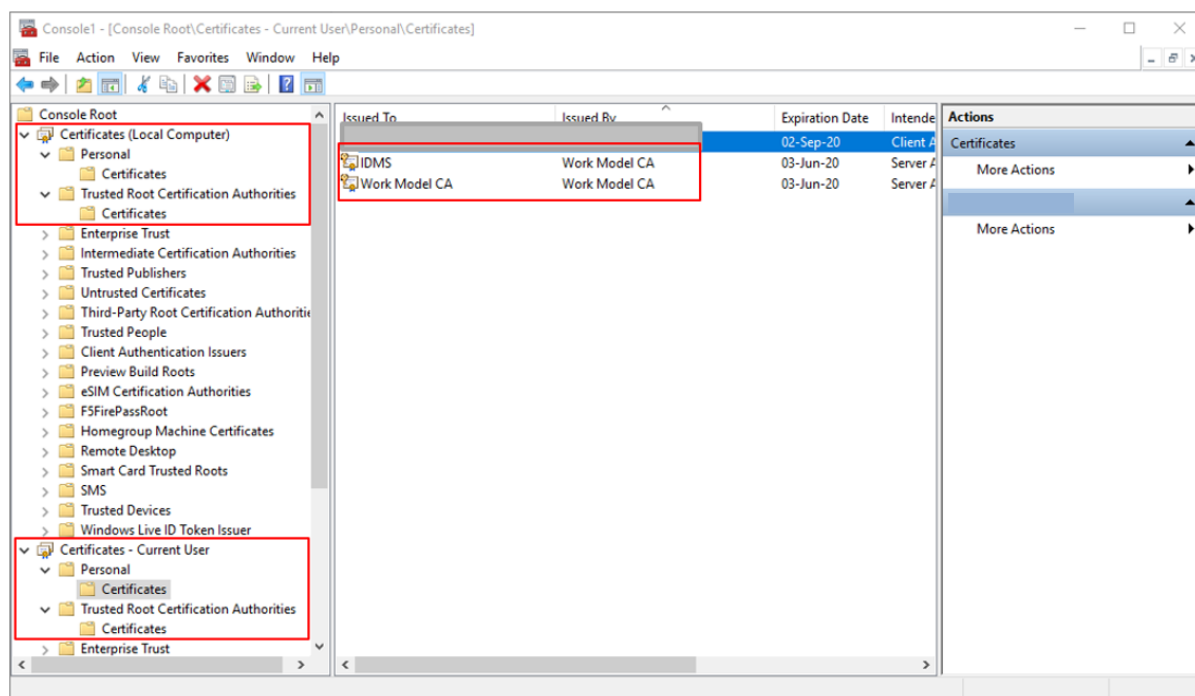
This allows access to the private key of the imported certificate to the user (service account) that is used to run ADMS servers. Without this, the servers cannot serve data in HTTPS mode.

22. Copy the Work Model Root CA and ADMS certificates using the context menu.
23. Navigate to the Certificates > Trusted Root Certification Authorities > Certificates folder.

24. Paste the ADMS and Work Model Root CA certificates from the Certificates folder context menu.



25. Add or copy the ADMS and Work Model Root CA certificates to the Certificates - Current User store (you may need to create the Current User store similar to the Certificates (Local Computer) store).



4.3. Configuring ADMS Components

! Important

When the Enterprise OMS system is deployed in Windows, you must specify the host and port values in the endpoint settings. When deployed in Kubernetes, you only need to specify the

host value.

Configuring the NetServerOmsConfig_localhost.txt file

To configure the %IDMS_ROOT%\config\server\NetServerOmsConfig_localhost.txt file:

1. Enable the Enterprise OMS database web service:

```
WORKMODEL_WEBDB_ENABLED = 1
```

2. Specify the Work Model Event Handler Service endpoint:

When deploying in Windows:

```
WORKMODEL_SERVICE_API = https://<IP of Work Model Event Handler Service host>:8080/
```

When deploying in Kubernetes:

```
WORKMODEL_SERVICE_API = https://<IP of Work Model Event Handler Service host>/
```

3. Specify the Work Model Notification Service events endpoint:

When deploying in Windows:

```
WORKMODEL_EVENT_API = ws://<IP of Work Model Notification Service host>:8083/oms/api/v1/events
```

When deploying in Kubernetes:

```
WORKMODEL_EVENT_API = ws://<IP of Work Model Notification Service host>/oms/api/v1/events
```

4. Specify Work Model Event Handler Service HTTPS parameters:

```
WORKMODEL_TLS_ENABLED = 1  
WORKMODEL_TLS_SUBJECTNAME = IDMS  
WORKMODEL_TLS_IGNORE_REMOTENAME_MISMATCH = 0
```

5. To enable completing Open and Close tasks that do not affect the customer energization status (for example, opening a device where downstream customers are already part of another incident), set the following:

```
OMS_CREATE_INCIDENTS_WHEN_NO_CUST_OUT = 1
```

Configuring the OmsDisplayPluginConfig.xml file

To configure the %IDMS_ROOT%\config\client\OmsDisplayPluginConfig.xml file:

1. Specify the Work Model Event Handler Service endpoint:

When deploying in Windows:


```
<WorkModelServiceApi>https://<IP of Work Model Event Handler Service  
host>:8080</WorkModelServiceApi>
```

When deploying in Kubernetes:

```
<WorkModelServiceApi>https://<IP of Work Model Event Handler Service host></WorkModelServiceApi>
```

2. Specify the Work Model Event Handler Service web socket endpoint:

When deploying in Windows:

```
<WorkModelWebsocketUrl>ws://<IP of Work Model Event Handler Service  
host>:8080/workModelClient</WorkModelWebsocketUrl>
```

When deploying in Kubernetes:

```
<WorkModelWebsocketUrl>ws://<IP of Work Model Event Handler Service  
host>/workModelClient</WorkModelWebsocketUrl>
```

3. Specify Work Model Event Handler Service HTTPS parameters:

```
<WorkModelTlsEnabled>true</WorkModelTlsEnabled>  
<WorkModelTlsSubjectName>IDMS</WorkModelTlsSubjectName>  
<WorkModelTlsIgnoreRemoteNameMismatch>>false</WorkModelTlsIgnoreRemoteNameMismatch>
```

Configuring the OmsInterfaceServerConfig.xml file

To configure the %IDMS_ROOT%\config\interfaces\OmsInterfaceServerConfig.xml file:

- Disable the User Account and Authentication (UAA) feature if you do not have a UAA server set up:

```
<UseUAAToken>>false</UseUAAToken>  
<UaaEndpoints></UaaEndpoints>  
<UseUAATokenForIntrospect>>false</UseUAATokenForIntrospect>  
<UaaIntrospectEndPoint></UaaIntrospectEndPoint>
```

Configuring the WorkOrderAdapterConfig.xml file

To configure the %IDMS_ROOT%\config\interfaces\WorkOrderAdapterConfig.xml file:

1. Specify the Work Model Event Handler Service's API endpoint:

When deploying in Windows:

```
<WorkOrderServiceUrl>https://<IP of Work Model Event Handler Service  
host>:8080/oms/api/v1/</WorkOrderServiceUrl>
```

When deploying in Kubernetes:

```
<WorkOrderServiceUrl>https://<IP of Work Model Event Handler Service
```

```
host>/oms/api/v1/</WorkOrderServiceUrl>
```

2. Specify Work Model Event Handler Service HTTPS parameters:

```
<WorkOrderTlsEnabled>>false</WorkOrderTlsEnabled>  
<WorkModelTlsIgnoreRemoteNameMismatch>>false</WorkModelTlsIgnoreRemoteNameMismatch>  
<WorkOrderTlsSubjectName>IDMS</WorkOrderTlsSubjectName>
```

3. Specify the path to the NetServerOmsConfig.txt file. Work Order Adapter reads the OMS_AUTOCLEAR_MOMENTARY_INCIDENTS parameter value. If auto clearance is disabled, the Work Order Adapter creates momentary orders for pending clear momentary incidents.

```
<NetServerOmsConfig>%IDMS_ROOT%\config\server\NetServerOmsConfig_localhost.txt</NetServerOmsConfig>
```

For more information about WorkOrderAdapterConfig.xml, see [WorkOrderAdapterConfig.xml](#).

Configuring the WorkOrderArchiveAdapterConfig.xml file

To configure the %IDMS_ROOT%\config\interfaces\WorkOrderArchiveAdapterConfig.xml file:

1. Specify the Work Model Notification Service events endpoint:

When deploying in Windows:

```
<WMNotificationServer>ws://<IP of Work Model Notification Service  
host>:8083/oms/api/v1/events</WMNotificationServer>
```

When deploying in Kubernetes:

```
<WMNotificationServer>ws://<IP of Work Model Notification Service  
host>/oms/api/v1/events</WMNotificationServer>
```

2. Specify Work Model Notification Service HTTPS parameters:

```
<WorkOrderTlsEnabled>>false</WorkOrderTlsEnabled>  
<WorkModelTlsIgnoreRemoteNameMismatch>>false</WorkModelTlsIgnoreRemoteNameMismatch>  
<WorkOrderTlsSubjectName>IDMS</WorkOrderTlsSubjectName>
```

3. Specify the path to the folder where Work Model journal files will be written to.

```
<WMJournalFilePath>%IDMS_ROOT%\dynamics\server\WMArchiveAdapter</WMJournalFilePath>
```

Configuring the WorkOrderDBSyncConfig.xml file

To configure the %IDMS_ROOT%\config\interfaces\WorkOrderDBSyncConfig.xml file:

1. Specify the path to the folder where Work Model journal files are written to.

```
<WMJournalFilePath>%IDMS_ROOT%\dynamics\server\WMArchiveAdapter</WMJournalFilePath>
```

2. Specify Enterprise OMS database connection parameters:

```
<WorkModelDBConfig>
<Name>DB1</Name>
<ConnectionString>Server=127.0.0.1;Port=5434;User
Id=postgres;Password=admin123;Database=workModel;Pooling=true;Minimum Pool Size=5;Maximum Pool
Size=100;</ConnectionString>
```

Configuring the ArchiveAppConfig.xml file

To configure the %IDMS_ROOT%\config\server\ArchiveAppConfig.xml file:

1. Enable writing the Work Model journal data to the ADMS Archive database:

```
<ArchiveWorkModelData>true</ArchiveWorkModelData>ArchiveWorkModelDataLogFilePathBase>
```

2. Specify the path to the folder where Work Model journal files are written to.

```
<WMJournalFilePath>%IDMS_ROOT%\dynamics\server\WMArchiveAdapter</WMJournalFilePath>
```

3. Specify the path to the folder where archived Work Model journal files are moved to.

```
<WorkModelDataArchivedPath>%IDMS_ROOT%\dynamics\server\WorkModel\Archived\</WorkModelDataArchivedP
ath>
```

Configuring the DataCorrectionInterfaceConfig.xml file

To configure the %IDMS_ROOT%\config\interfaces\DataCorrectionInterfaceConfig.xml file:

1. Specify the GraphQL service port number the Data Correction Interface application runs on:

```
<GraphQLPort>50001</GraphQLPort>
```

2. Specify the Work Model Event Handler service REST API URL to get cancelled and closed orders:

When deploying in Windows:

```
<WorkOrderServiceUrl>http://<IP of Work Model Event Handler service
host>:8080/oms/api/v1/</WorkOrderServiceUrl>
```

When deploying in Kubernetes:

```
<WorkOrderServiceUrl>http://<IP of Work Model Event Handler service
host>/oms/api/v1/</WorkOrderServiceUrl>
```

3. Specify Work Model Event Handler Service HTTPS parameters:

```
<WorkOrderTlsEnabled>false</WorkOrderTlsEnabled>
<WorkModelTlsIgnoreRemoteNameMismatch>false</WorkModelTlsIgnoreRemoteNameMismatch>
<WorkOrderTlsSubjectName>IDMS</WorkOrderTlsSubjectName>
```

4. Specify the Net Data Server web API URL to get cancelled and cleared incidents:

```
<DataCorrectionIncidentsHostURL>http://<IP of Net Data Server  
host>:8443/api/Oms/v2/Oms</DataCorrectionIncidentsHostURL>
```

For more information about the Data Correction Interface application, refer to the *EOMS Data Correction Interface User Guide*.

Configuring the ADMS Client Shortcut

Add the -CLIENTTYPE "WORKMODELDEMO" parameter to the ADMS client shortcut that starts the ADMS client (for example, ADMS-APF Client-Localhost.lnk). This will show additional Enterprise OMS controls and data panels for users as specified in the client appearance configuration file, ViewerConfig.xml.

Configuring Order Name Format and Order Custom Fields

You can customize the order name format and define order custom fields using the OMS Client Configuration Editor. For more details, refer to the *OMS Client Configuration Editor User Guide*.

Configuring Coded Translation Items

You can customize the order-related, coded translation items (for example, WorkOrderSuspendedReasons, which lists reasons for suspending an order) in the Coded Translation Editor. For more details, refer to the *Coded Translation Editor User Guide*.

Configuring Enterprise OMS Organizations for OMS

For more information about configuring the system for the organizations functionality, see [Enterprise OMS Organizations Definition](#).

Showing Order Symbols on the Network View

1. Customize the order symbols on the network view for different conditions, such as order status or assigned crew status using the OMS Server Configuration Editor. For more details, refer to the *OMS Server Configuration Editor User Guide*.
2. Edit the %IDMS_ROOT%\config\client\netconf_localhost.xml file as follows:

```
<ShowOrdersInsteadOfIncidents>true</ShowOrdersInsteadOfIncidents>
```

This setting defines that order symbols appear instead of incident symbols on the network view.

Setting the auro configuration parameters for Planned Switching orders, Manual Switching orders, and Load shed orders

The behavior of automatically created Planned Switching, Manual Switching, and Loadshed orders can be configured using the parameters: ORDERS_AUTOCONFIG_PLANNEDSWITCHING, ORDERS_AUTOCONFIG_MANUALSWITCHING, and ORDERS_AUTOCONFIG_LOADSHED in the OMS Workmodel properties file. These parameters determine the behavior of such orders when all customers have been restored. When the configuration parameter for an order type is set to:

- **0:** Orders will remain "In Progress" upon restoration.
- **1:** Orders will automatically transition to "Field Complete" on restoration and remain in this state.
- **2:** Orders will be transitioned through "Field Complete" to "Closed", with the associated incident also being cleared.

Note

These automatic status transitions may not occur if there are constraints that prevent them, such as the presence of incomplete tasks, missing mandatory fields, or failed validation rules.

4.4. Configuring ADMS Components for a Multi-Host System

This section describes a multi-host system that supports EOMS high availability (also referred to as redundancy or failover). This configuration applies to Kubernetes and Windows installations. For more information about configuring EOMS components, see [Configuring EOMS Components for a Multi-Host System](#).

Prerequisites for EOMS multi-host system are as follows:

- ADMS multi-host system is configured for Active and Standby modes. For more information, refer to the *ADMS Installation Guide*.
- Redundant ADMS and EOMS servers have similar configurations and have File Replicator and MRS services running on ADMS servers.

For example, four servers are running (ADMS1, ADMS2, EOMS1, and EOMS2). ADMS1 is "Enabled" and ADMS2 is "Standby"; EOMS1 is "Enabled" and EOMS2 is "Standby".

An EOMS high availability system is available through the ADMS "Active" and "Standby" system configuration. The following applications support an EOMS high availability system:

- Configuring the Work Order Archive Adapter

This application listens for the events from Work Model Notifications service and writes Work Model journal files. For more information, see [Enterprise OMS Journaling and Archiving](#).

For Work Order Archive Adapter, ADMS1 is connected to EOMS1, and ADMS2 is connected to EOMS2.

Example configuration:

```
<WorkModelArchiveAdapterconfig>
  <WMNotificationServer>wss://EOMS1:8083/oms/api/v1/events</WMNotificationServer>
  <WorkModelTlsEnabled>true</WorkModelTlsEnabled>
  <WorkModelTlsSubjectName>"Certificate Subject Name"</WorkModelTlsSubjectName>
  ...
</WorkModelArchiveAdapterconfig>
```

```
</WorkModelAchiveAdapterconfig>
```

- Configuring the Work Order Adapter

This application reads the Work Model journal files and sends the journals to work-model-eventhandler to be processed by work-model-service.

Work Order Adapter supports a many-to-many relationship between EOMS and ADMS. You can define multiple comma-separated EOMS REST endpoint entries using the WorkOrderServiceUrl parameter in the WorkOrderAdapterConfig.xml configuration file. Multiple Work Order Adapter instances in a high availability (HA) system can use similar configuration files and switch to the active EOMS REST endpoint at the initialization or runtime.

Example configuration:

```
<WorkOrderAdapterConfig>
  <WorkOrderServiceUrl>https://EOMS:8080/oms/api/v1/,
https://EOMS2:8080/oms/api/v1/</WorkOrderServiceUrl>
  <WorkOrderTlsEnabled>true</WorkOrderTlsEnabled>
  <WorkModelTlsIgnoreRemoteNameMismatch>true</WorkModelTlsIgnoreRemoteNameMismatch>
  <WorkOrderTlsSubjectName>IDMS</WorkOrderTlsSubjectName>
  ...
</WorkOrderAdapterConfig>
```

- Configuring the Work Order Database Sync

This application reads Work Model journal files and writes the data to the backup Enterprise OMS database.

For Work Order Database Sync, ADMS1 is connected to EOMS2, and ADMS2 is connected to EOMS1.

Example configuration:

```
<ConnectionString>Server=EOMS;Port=5432;User
Id=postgres;Password=admin1;Database=workModel;Pooling=true;Minimum Pool Size=5;Maximum Pool
Size=100</ConnectionString>
```

- Configuring the NetServer (OMS Server):

```
WORKMODEL_WEBDB_ENABLED=1
WORKMODEL_SERVICE_API = https://EOMS1:8080/,https://EOMS2:8080/
WORKMODEL_EVENT_API = wss://EOMS1:8083/oms/api/v1/events, wss://EOMS2:8083/oms/api/v1/events
' Enable HTTPS to the Work Model (1 = Enable, 0 = Disable)
WORKMODEL_TLS_ENABLED=1
' The partial string to find the Client Certificate to send from the Personal Certificate Storage.
WORKMODEL_TLS_SUBJECTNAME = IDMS
' (1 = Ignore the SubjAltName being different than the hostname that is running the work model, 0
= Validate Certificates Fully)
WORKMODEL_TLS_IGNORE_REMOTENAME_MISMATCH=1
```

- Configuring the ADMS client:

For APF clients to failover, add EOMS event URLs to OmsDisplayPluginConfig.xml (by default, stored at %IDMS_ROOT%\config\client\)

```
<WorkModelServiceApi>https://EOMS1:8080,https://EOMS2:8080</WorkModelServiceApi>
<WorkModelTlsSubjectName>IDMS</WorkModelTlsSubjectName>
<WorkModelTlsEnabled>true</WorkModelTlsEnabled>
<WorkModelTlsIgnoreRemoteNameMismatch>true</WorkModelTlsIgnoreRemoteNameMismatch>
<WorkModelWebsocketUrl>wss://EOMS1:8080/workModelClient,wss://EOMS2:8080/workModelClient</WorkModelWebsocketUrl>
<WorkModelWSRequestTimeout>60</WorkModelWSRequestTimeout>
```

The following diagram shows how the Work Model journal files update the backup Enterprise OMS database in a high availability system to have the primary and backup Enterprise OMS databases in sync.

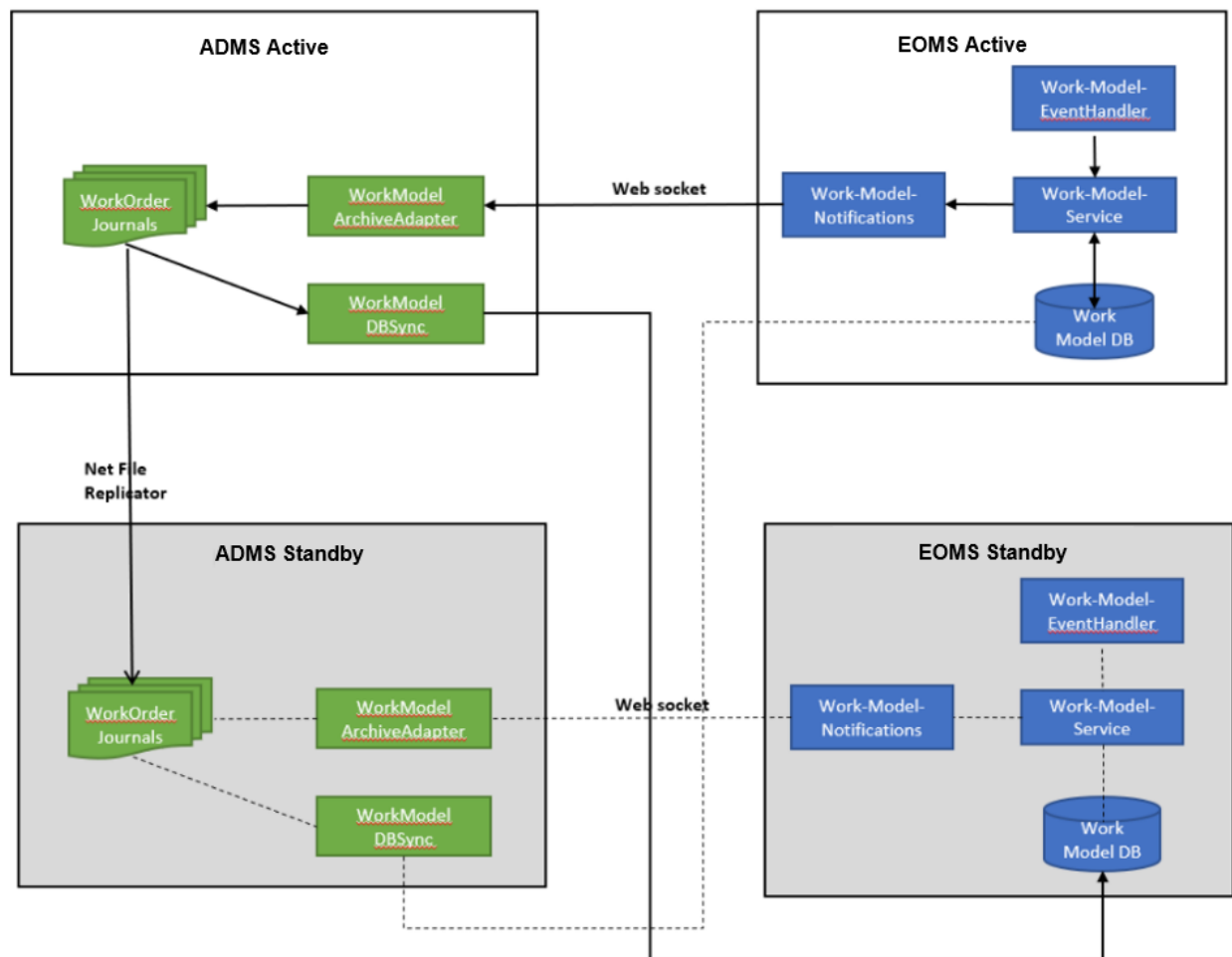


Figure 7. Enterprise OMS High Availability System

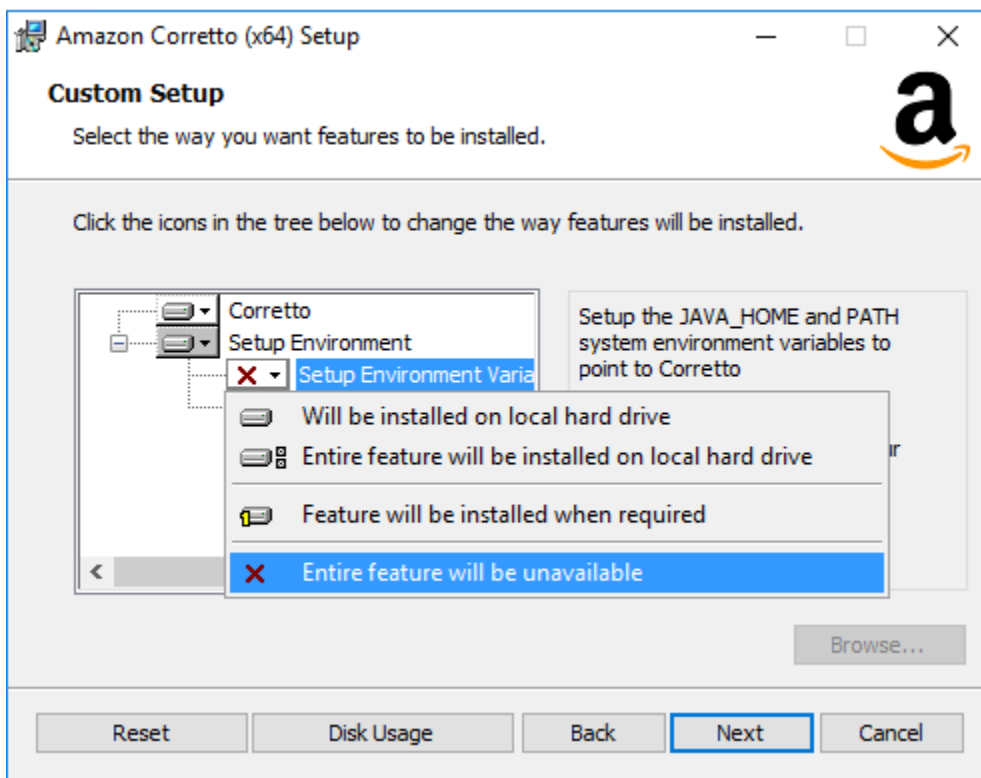
5. Installing Enterprise OMS

5.1. Installing Prerequisites

Amazon Corretto is required to run the Enterprise OMS service Java executable files.

To install Amazon Corretto:

1. Download amazon-corretto-17.0.8.8.1-windows-x64.msi from <https://github.com/corretto/corretto-11/releases>.
2. Run the installer.
3. On the Custom Setup parameters page, choose the "Entire feature will be unavailable" option for the "Setup Environment Variables" feature.



4. Update the installation path so that the installation folder path and name do not include spaces (for example, install to C:\Corretto).
5. Install the application.
6. Add the CORRETTO_HOME user environment variable with the path "<Corretto_Installation_Folder>\jdk11.0.8_10\bin". For example:

```
setx CORRETTO_HOME C:\Corretto\jdk11.0.8_10\bin
```



Important

PROCMAN cannot resolve CORRETTO_HOME paths with spaces.

5.2. Installing in Kubernetes

Note

All machines must be in one network and must be able to ping each other and exchange data requests and responses.

To deploy Enterprise OMS:

1. Navigate to <Enterprise_OMS_Deployment_Folder>.
2. Execute the following commands:

```
bsfdeploy --action apply -i foundation-manifest.yaml
bsfdeploy --action apply -i wm-manifest.yaml
```

These commands are used to deploy Foundation Base and Enterprise OMS using the Bsfdeploy client. For the Bsfdeploy installation and usage details, refer to the *Bsfdeploy User Guide*.

To verify the Enterprise OMS deployment:

1. Execute the following command:

```
helm list
```

2. Verify that the following Enterprise OMS components are deployed.

```
work-model-schema-job
work-model-service
work-model-eventhandler
work-model-notifications
```

3. Execute the following command:

```
kubectl get pods --namespace [.mark]#work-model#
```

4. Verify that Enterprise OMS containers are listed. It may take several minutes for the containers to start up.

☐ Name ▾	Namespace ▾
☐ work-model-eventhandler-5f85df996f-4ck...	work-model
☐ work-model-notifications-75ccc4cc6f-nvq...	work-model
☐ work-model-ois-service-7cbcfc87c9-grnsh	work-model
☐ work-model-service-7d5b58bf79-2sljg	work-model
☐ work-model-service-7d5b58bf79-gtv5r	work-model

5. Run the Work Order Adapter.
6. Verify that successful connection messages are shown.

5.3. Installing in Windows

Note

It is assumed that you already have the ADMS server, client, and interfaces installed.

To install Enterprise OMS services:

1. Run the ADMS_Enterprise_OMS_Module.msi installer.
2. Install the Enterprise OMS using default installation options.

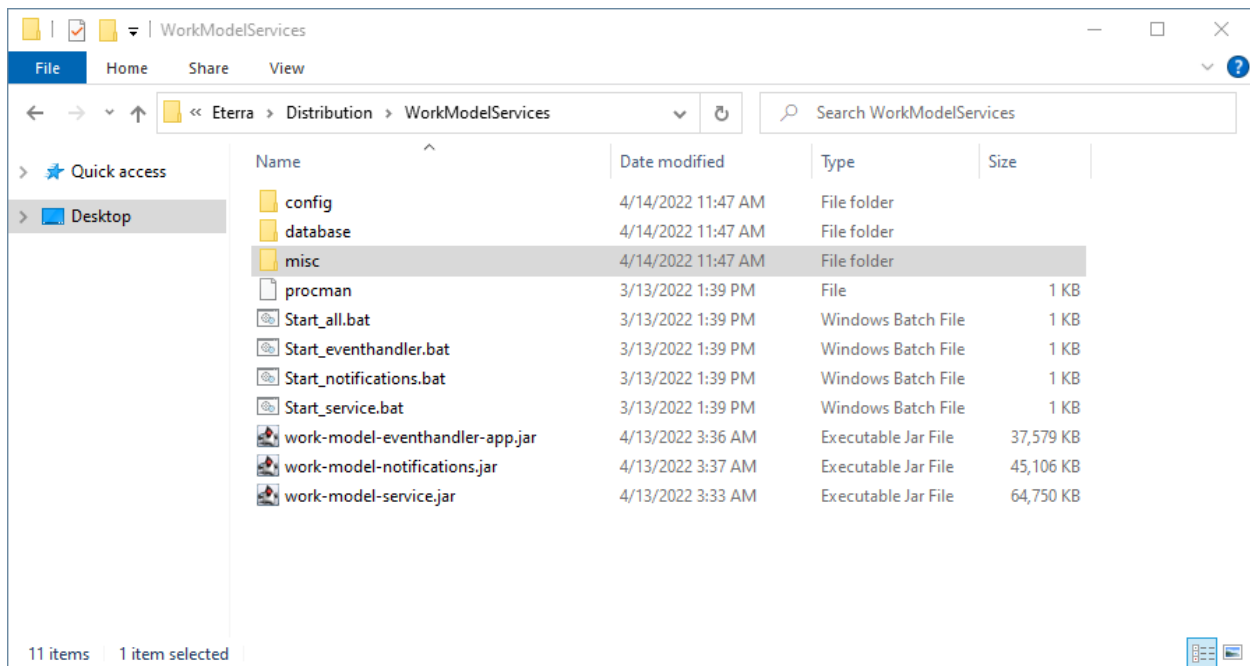


Figure 8. Enterprise OMS Services Folder

By default, Enterprise OMS services are installed to the Distribution/WorkModelServices folder (also referred to as the Enterprise OMS installation folder). The installation includes the following files:

- Configuration files for Enterprise OMS services.
- Scripts to deploy the Enterprise OMS database.
- Application definition file and PROCMAN model to start Enterprise OMS services under the PROCMAN.
- Java executables for Enterprise OMS services.

5.3.1. Setting Up the Database

1. Download a PostgreSQL installer from <https://www.postgresql.org/download/windows/>.
2. Install PostgreSQL.
 - Leave default values for the port and username (Port = 5432, Username = postgres).
 - Set Password = admin1.
3. Copy the Enterprise OMS database installation script psql_dump.sql from the "database" folder under the Enterprise OMS Installation Folder to the PostgreSQL bin folder (for example, C:\Program Files\PostgreSQL\14\bin).
4. In command line, navigate to the PostgreSQL bin folder.
5. Execute the following command to create the workModel database, and enter the workModel database password:

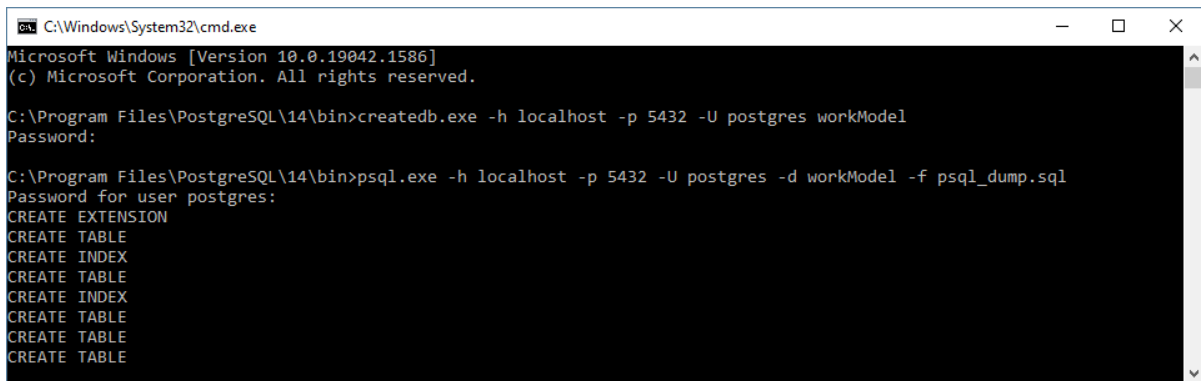
```
createdb.exe -h localhost -p 5432 -U postgres workModel
```

Note

This command assumes that you use the default PostgreSQL server host name, port, and user.

6. Execute the following command to create the workModel database schema:

```
psql.exe -h localhost -p 5432 -U postgres -d workModel -f psql_dump.sql
```



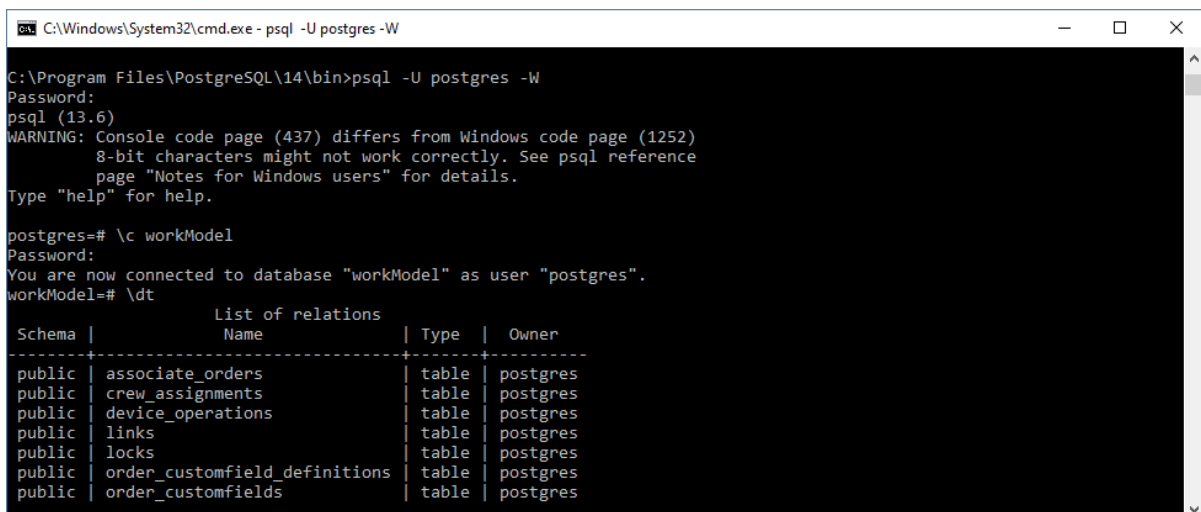
```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19042.1586]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files\PostgreSQL\14\bin>createdb.exe -h localhost -p 5432 -U postgres workModel
Password:

C:\Program Files\PostgreSQL\14\bin>psql.exe -h localhost -p 5432 -U postgres -d workModel -f psql_dump.sql
Password for user postgres:
CREATE EXTENSION
CREATE TABLE
CREATE INDEX
CREATE TABLE
CREATE INDEX
CREATE TABLE
CREATE TABLE
CREATE TABLE
```

7. Execute the following commands to get the list of tables in the workModel database:

```
psql -U postgres -W
postgres=# \c workModel
workModel=# \dt
```



```
C:\Windows\System32\cmd.exe - psql -U postgres -W

C:\Program Files\PostgreSQL\14\bin>psql -U postgres -W
Password:
psql (13.6)
WARNING: Console code page (437) differs from Windows code page (1252)
8-bit characters might not work correctly. See psql reference
page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \c workModel
Password:
You are now connected to database "workModel" as user "postgres".
workModel=# \dt

      List of relations
Schema | Name | Type | Owner
-----+-----+-----+-----
public | associate_orders | table | postgres
public | crew_assignments | table | postgres
public | device_operations | table | postgres
public | links | table | postgres
public | locks | table | postgres
public | order_customfield_definitions | table | postgres
public | order_customfields | table | postgres
```

8. Make sure Enterprise OMS tables appear:

- associate_orders
- crew_assignments

- device_operations
- ...

5.3.2. Configuring Database SSL Configuration

Important

When configuring PostgreSQL to use SSL, the "root.crt" certificate must be the same CA certificate that was used to sign the Enterprise OMS services certificates. All certificates must share a trusted CA.

1. Edit the pg_hba.conf file in %PGDATA% as follows:

Existing Setting:

```
host all all all md5
```

New Setting (Force SSL Only):

```
hostssl all all all cert clientcert=1
```

Note

If you want to enable both SSL and Non-SSL connections, do not change the file pg_hba.conf. The default settings will allow both if "ssl = on" in postgresql.conf.

2. Edit the postgresql.conf file in %PGDATA% as follows:

Uncomment SSL settings. Set SSL to enabled (=on) and update SSL paths with the paths to the certificate files that you generated for the Enterprise OMS. For example:

```
ssl = on
ssl_ca_file = 'D:\Eterra\Distribution\WorkModelServices\root.crt'
ssl_cert_file = 'D:\Eterra\Distribution\WorkModelServices\server.crt'
ssl_key_file = 'D:\Eterra\Distribution\WorkModelServices\server.key'
```

Note

The file root.crt is the same exact file as ca.pem but with a different name. Do not change anything else except the name of the file.

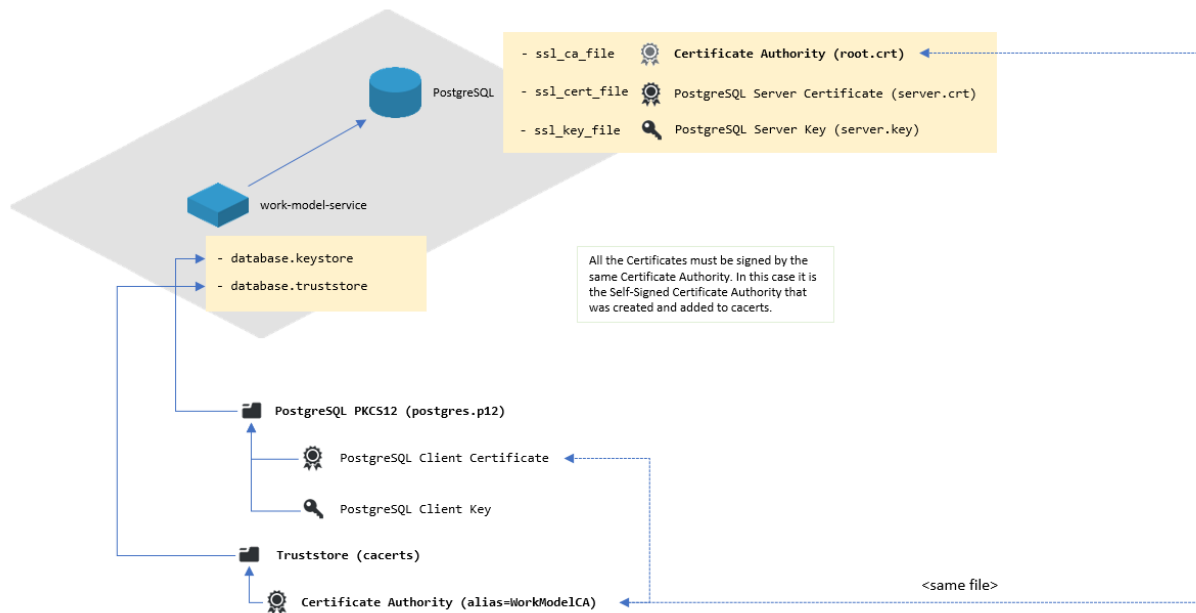


Figure 9. PostgreSQL SSL Configuration

5.3.3. Upgrading Work Model Database to a Newer Version

To upgrade the Work Model database to a newer version (for example, from EOMS 1.4 to EOMS 1.5):

1. Copy the Enterprise OMS database upgrade script `psql_upgrade.sql` from the "database" folder under the Enterprise OMS Installation Folder to the PostgreSQL bin folder (for example, `C:\Program Files\PostgreSQL\14\bin`).
2. In command line, navigate to the PostgreSQL bin folder.
3. Execute the following command:

```
psql.exe -h localhost -p 5432 -U postgres -d workModel -f psql_upgrade.sql
```

5.3.4. Upgrading PostgreSQL to a Newer Version

To upgrade PostgreSQL to a newer version (for example, from 12.5 to 14.1):

1. Download a PostgreSQL installer from <https://www.postgresql.org/download/windows/>.
2. In command line, navigate to the PostgreSQL bin folder.
3. Execute the following command:

```
pg_dump -U postgres workModel > c:\temp\WorkModelDbBackup.sql
```

This backs up the existing Work Model database to the `WorkModelDB.sql` file.

4. Stop the running database instance.
5. Install a new PostgreSQL instance (use the default port 5432).
6. Start the new PostgreSQL instance.
7. In command line, navigate to the PostgreSQL bin folder.
8. Execute the following command:

```
createdb.exe -h localhost -p 5432 -U postgres workModel
```

This creates the Work Model database.

9. Execute the following command:

```
psql -h localhost -p 5432 -U postgres workModel < c:\temp\WorkModelDbBackup.sql
```

This restores the Work Model database from the backup.

6. Advanced Configuration

6.1. Configuring Enterprise OMS Services

The configuration parameters for each service can be set using environment variables.

6.1.1. Parameters for Every Service

- **SERVER_PORT**: The port on which to run the service.
- **SSL connection handle** by foundation component using the gatewayroute which uses the ingress authentication of each services.

Note

The following interval-based parameters require specific time durations, such as the number of days, minutes, or seconds to be specified:

EVENTS_CLEAN_AGE

EVENTS_CLEAN_INITIAL_DELAY

EVENTS_CLEAN_DELAY

REMOVEORDERS_REMOVE_INITIAL_DELAY

REMOVEORDERS_REMOVE_DELAY

REMOVEORDERS_REMOVE_RETENTION_PERIOD

Month- or year-based intervals are not supported for these parameters.

- **EVENTS_CLEAN_AGE**: An ISO-8601 duration that determines how old events can be before they are cleaned.

The default value is default "PT10M". This value should be changed to a value appropriate for production deployment.

- **EVENTS_CLEAN_INITIAL_DELAY**: An ISO-8601 duration that determines the delay before the very first event clean check.

The default value is "PT1M".

- **EVENTS_CLEAN_DELAY**: An ISO-8601 duration that determines the time between the end of one clean task and the start of the next.

The default value is "PT10M".

- **HAZELCAST_PORT**: The port that Hazelcast will bind to.

The default value is "5701".

- **K8S_SERVICE_NAMESPACE**: The Kubernetes namespace this service is being deployed in.

- `K8S_SERVICE_HAZELCASTPORT`: The port that Hazelcast is running on. This does not change the port, only for connections.
- `REMOVEORDERS_REMOVE_INITIAL_DELAY`: The duration of time after the service first starts before it searches for closed or cancelled orders that can be removed.
- `REMOVEORDERS_REMOVE_DELAY`: The duration of time the service waits in between each subsequent attempt to remove old orders.
- `REMOVEORDERS_REMOVE_RETENTION_PERIOD`: This is how old an order has to be before it can be removed.

6.1.2. Parameters for work-model-notifications

- `SERVER_PORT`: The port on which to run the service.
- `K8S_SERVICE_NAMESPACE`: The Kubernetes namespace this service is being deployed in.
- `K8S_SERVICE_HAZELCASTPORT`: The port that Hazelcast is running on. This does not change the port, only for connections.
- `KEY`: The base64 encrypted AES key that is used to encrypt data for the notification system.
- `KEY_ALGORITHM`: The algorithm used to encrypt data for the notification system.

The default value is "AES".

- `HAZELCAST_CLIENT_ADDRESSES`: The locations of the work-model-service Hazelcast ports when running in standalone mode. Example: localhost:5701.

6.1.3. Parameters for work-model-eventhandler

- `SERVER_PORT`: The port on which to run the service.

The default value is "false". Do not set this to "true" unless you understand the implications of doing so.

- `SERVICES_WORKMODELSERVICE_URL`: The URL to access the Work Model Service. Example: <https://work-model-service>.
- `SERVICES_OMSINTERFACESERVER_URL`: The URL to access the OMS Interface Server.
- `ORDERS_AUTOCONFIG_MANUALSWITCHING`: Determines if the Work Model should automatically close this order once it is completed. Possible values: 0 = do nothing, 1 auto field complete, 2 auto close.
- `ORDERS_AUTOCONFIG_PLANNEDSWITCHING`: Determines if the Work Model should automatically close this order once it is completed. Possible values: 0 = do nothing, 1 auto field complete, 2 auto close.
- `ORDERS_AUTOCONFIG_LOADSHED`: Determines if the Work Model should automatically close this order once it is completed. Possible values: 0 = do nothing, 1 auto field complete, 2 auto close.

- LOGGING.LEVEL.ROOT: Default logging level equal to ERROR logging.
- LOGGING.LEVEL.SPRING: Logging level parameter for Spring library.

The default logging levels are set to "ERROR" as the logs can be very verbose.

- LOGGING.LEVEL.COM.GE: Logging level parameter for official company notification.
- LOGGING.LEVEL.DATA: Logging level parameter for Data Service.
- LOGGING.LEVEL.ETERRA: Logging level parameter for Eterra Service.

The default logging levels value are set to "INFO" as an operational level. To increase the log levels and run diagnostics, switch parameter value to "DEBUG".

7. Running Enterprise OMS

To start Enterprise OMS:

1. Run ADMS servers, including the NetServer (OMS Server).
2. Run ADMS interface servers, including the OMS Interface Server, Work Order Adapter, Work Order Archive Adapter, Work Order Database Sync, Net Data Server, and Data Correction Interface.
3. Run the ADMS client.
4. Run the Enterprise OMS services. See the following sections for details.

The Enterprise OMS application is now ready to use.

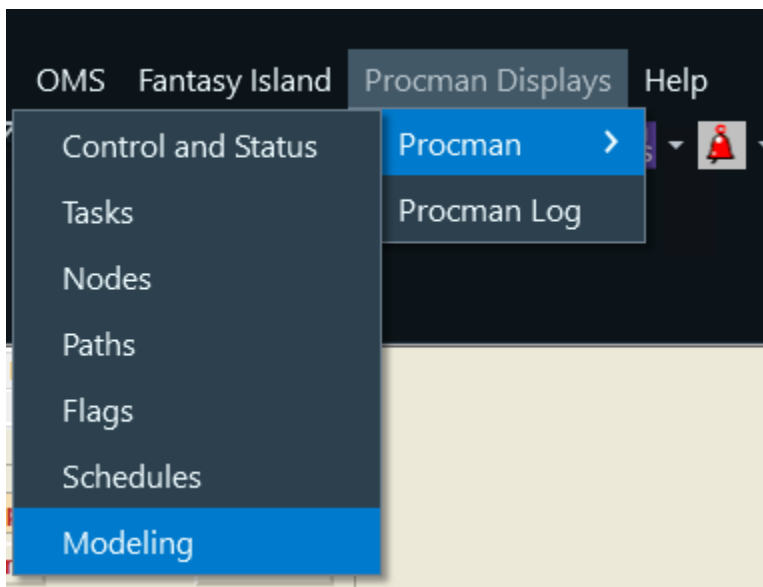
7.1. Running ADMS Interface Servers

To run ADMS Interface Servers:

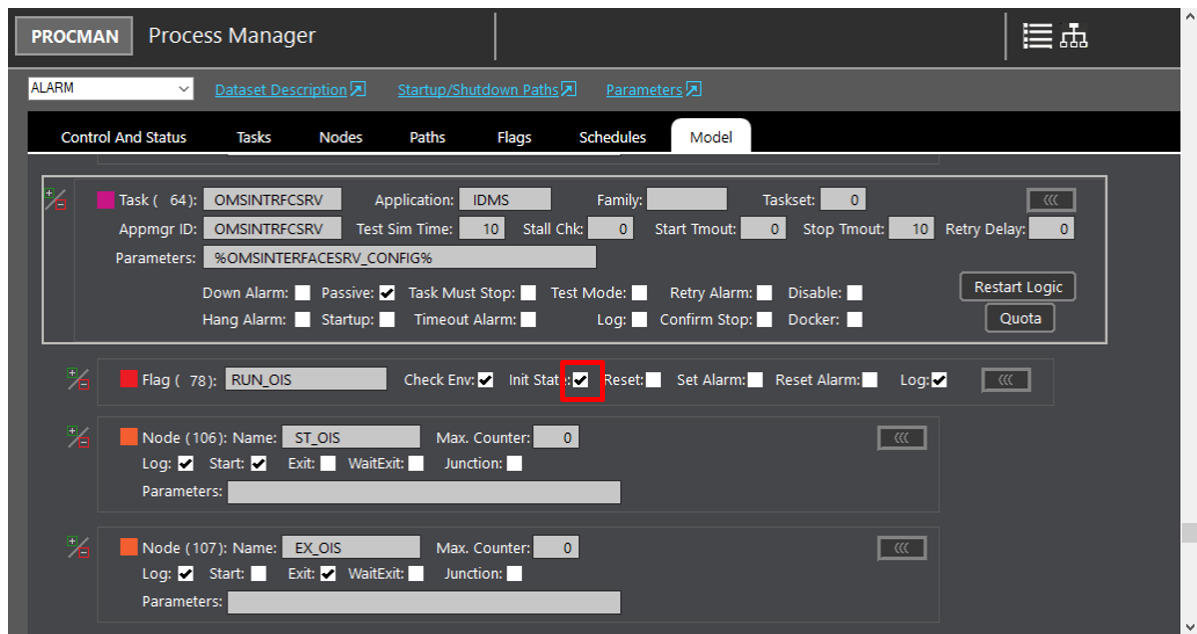
- Run ADMS interface servers (OMS Interface Server, Work Order Adapter, Work Order Archive Adapter, Work Order Database Sync, Net Data Server, and Data Correction Interface) via PROCMAN or the Windows Start menu.

To configure the ADMS interface server tasks to automatically start:

1. In the Browser client, click **HABITAT Applications > Process Manager > Process Manager Master**.
2. On the **Procman Displays** menu, click **Procman > Modeling**.



3. Set the **Init State** flag for the ADMS interface server tasks (OMSINTRFCSRV, NETDATASRV, DATACORR, WMARCHADAPT, WMADAPTER, WMDBSYNC).



Or:

1. Open a command prompt.
2. Execute the following commands:

```
setx PROCMAN_FLAG_EMS_RUN_NETDATASRV TRUE
setx PROCMAN_FLAG_EMS_RUN_OIS TRUE
setx PROCMAN_FLAG_EMS_RUN_DATACORR TRUE
setx PROCMAN_FLAG_EMS_RUN_WMADAPTER TRUE
setx PROCMAN_FLAG_EMS_RUN_WMARCHADAPT TRUE
setx PROCMAN_FLAG_EMS_RUN_WMDBSYNC TRUE
```

7.2. Running Enterprise OMS Services in Kubernetes

To run Enterprise OMS services in Kubernetes:

- Run Foundation and Enterprise OMS pods in Kubernetes.

7.3. Running Enterprise OMS Services in Windows

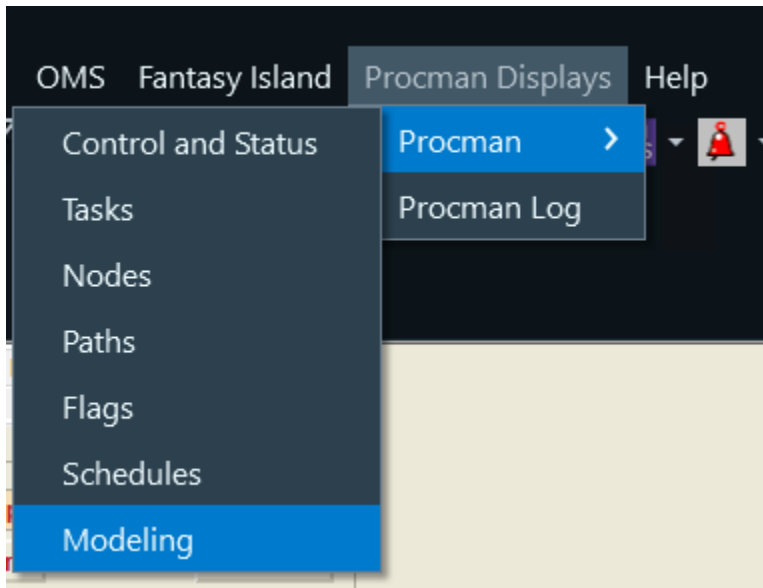
To run the Enterprise OMS services in Windows:

- Run the Enterprise OMS services (Event Handler, Notifications, and Service) via PROCMAN or Windows Start menu.

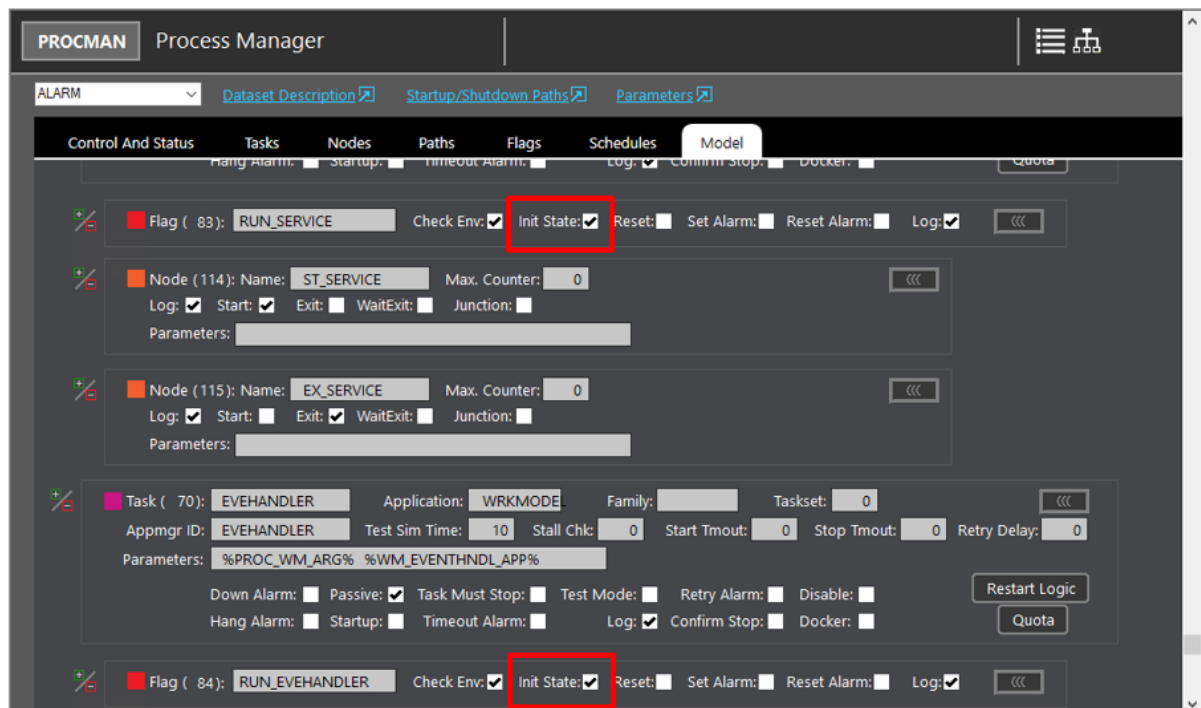
To configure the Enterprise OMS service tasks to automatically start:

1. In the Browser client, click **HABITAT Applications > Process Manager > Process Manager Master**.

- On the **Procman Displays** menu, click **Procman > Modeling**.



- Set the **Init State** flag for the Enterprise OMS service tasks (EVEHANDLER, NOTIFICATION, and SERVICE).



Or:

- Open a command prompt.
- Execute the following commands:

```
setx PROCMAN_FLAG_EMS_RUN_EVEHANDLER TRUE
```

```
setx PROCMAN_FLAG_EMS_RUN_SERVICE TRUE  
setx PROCMAN_FLAG_EMS_RUN_NOTIFIC TRUE
```

8. Maintenance and Troubleshooting

8.1. Verifying the Setup

To verify that you have correctly configured and run the system:

1. Run the ADMS system, Work Order Adapter, and Enterprise OMS components.
2. In ADMS, create an incident.
3. Check that the Orders tab in the OMS Display plug-in contains an order related to the incident.

8.2. Upgrading to a Newer Version

To upgrade to a newer version of Enterprise OMS using Helm, enter the following:

```
helm upgrade --namespace etd --name notify <service container namespace>
```

8.3. Maintenance

8.3.1. Database Schema

When the database schema changes, the database volume needs to be re-created in order to work with the proper schema. This can be done by removing all of the volumes attached to the contains via docker-compose.

8.3.2. Profiling

```
XX:StartFlightRecording=disk=true,\  
  dumponexit=true,filename=recording.jfr,\  
  maxsize=1024m,maxage=1d,settings=profile,\  
  path-to-gc-roots=true
```

8.3.3. Services

Enterprise OMS services:

- **work-model-schema-job:** This service executes the Postgres SQL database server. Running this container will automatically configure the database with the correct schema and database version.
- **work-model-service:** This service supports logic to access the database.

- **work-model-eventhandler:** This service supports the business logic to process events within the system.
- **work-model-notifications:** This service listens for updates to orders and sends them to ADMS.

8.3.4. Endpoints

Reference the individual service below to determine which endpoints they expose. Two services expose public endpoints:

```
work-model-eventhandler - The REST API on port 8080 by default.  
work-model-notifications - The Event Stream on port 8083 by default.
```

To invoke the Enterprise OMS API:

```
http://localhost:8080/oms/api/v1
```

To listen for events:

```
http://localhost:8083/oms/api/v1/events
```

9. Generating an SSL/TLS Certificate

Enterprise OMS uses Public Key Infrastructure to secure communications between all of the Enterprise OMS components, including the integration with the ADMS.

PKI requires the use of Public Certificates which are signed by a Trusted Certificate Authority. In order to sign the certificates, a Certificate Authority (CA) must be created first. Websites use known Certificate Authorities and request for their public Certificates to be signed via an application process. As this is an enterprise product, and the services will be used internally and not publicly, it is sufficient to create a "Self-Signed Certificate Authority."

9.1. Required Tools

- OpenSSL (for more information, see <https://wiki.openssl.org/index.php/Binaries>)

Windows: Install using Choco:

```
choco install openssl
```

Linux: Install using:

```
apt-get
```

- keytool: This is part of the JDK 11.

9.2. Self-Signed Certificate Authority

The self-signed CA is essentially a certificate which is signed and trusted at the same time. The self-signed CA is the root of the certificate chain. By default, no product or Java Virtual Machine will trust this Certificate Authority because it is not well-known. For the system to trust your own self-signed CA, the self-signed CA must be imported in the system's certificate store.

Create Self-Signed CA

The following command creates a random key that will be the private part of the Certificate Authority that you create.

Note

If a password is required, then add the option `-aes256`. For example, `openssl genrsa -aes256 -out service.key 2048`. The command will prompt for a password. Whenever this key is used, it will require the password, so you must remember it. Without it, no new certificates can be signed, and the rootCA not be usable.

```
# Create the root CA private key.
```

```
openssl genrsa -out rootCA.key 2048
```

The next step is to create a new Certificate and then sign that Certificate with itself, producing what is called a Self-Signed Certificate Authority.

```
# Create and Sign the Root Certificate Authority
openssl req -x509 -new -nodes -key rootCA.key -sha256 -days 3650 -out rootCA.crt -subj
"/C=US/ST=FL/O=GE Digital/CN=Work Model CA"
```

There are a few important parameters in this command:

- **-days 3650:** This is the number of days for which the Certificate Authority will be valid. This value causes all of the signed Certificates to also become invalid once the number of days has been reached. It is important for this value to be long enough so that it does not cause significant maintenance issues, but not too long as to be insecure.
- **-subj "":** This parameter automatically enters information for the certificate that would typically be asked on the command line.
 - **/C:** Country of Origin (2 Letter Code).
 - **/ST:** State of Origin (2 Letter Code).
 - **/O:** Organization (this value should match the same value for the service certificates).
 - **/CN:** Common Name. This value is the most important. For the Root Certificate Authority, it should always be "Work Model CA". However, when it is used for Service Certificates, it must match the hostname of the server the service will be running on.

9.3. Create Service Certificate

Create a private key that will be the counterpart to the Service Certificate:

This is the same process as described in the previous section. Each Certificate requires a private key. This will create a private key that is not protected with a password. The Keystore that this key and certificate will be imported into can be protected, so it is best not to password protect this key as it can cause difficulties later.

```
# Create the Private Key for the Service
openssl genrsa -out service.key 2048
```

Create a request for the private key.

This request is used to sign the new certificate with the previous `rootCA.crt` file. This contains the information that will be used to create the matching certificate for the Enterprise OMS Service that will use it. This certificate will be presented when the service connects to other services to prove that the service is authentic. As the certificate will be signed by the rootCA, and every other service will be signed by the same rootCA, the services will all trust each other.

```
# Create a request for the private key.
```

```
openssl req -new -sha256 -key service.key -subj "/C=US/ST=CA/O=GE Digital/CN=localhost" -out service.csr
```

This command has similar commands as the root CA one:

- **-subj ""**: This parameter automatically enters information for the certificate that would typically be asked on the command line.
- **/C**: Country of Origin (2 Letter Code).
- **/ST**: State of Origin (2 Letter Code).
- **/O**: Organization (this value should match the same value for the service certificates).
- **/CN**: Common Name. This value is the most important. It must match the hostname of the server that the service will be running on. If this is being used locally for development, then localhost is a good choice. If it is being run on production, or in k8s, then the hostname of the server or container should be used.

Create the new Certificate for the service and sign it with the Root Certificate Authority.

This process requires access to the Root CA Certificate *and* Root CA Private Key. If a password was put on the Root CA Private Key, it will prompt you for that password.

```
# Create and Sign the new Service Certificate
openssl x509 -req -in service.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out service.crt
-days 3650 -sha256
```

The most important dynamic parameter is the days parameter:

- **-days 3650**: This is the number of days for which the Certificate will be valid. Certificates cannot be extended; once this value expires, a new Certificate must be created. This value must be the same or less than the Root CA value. If it is set to a greater value, it will not make a difference, since the Root Certificate Authority will have already expired.

Create a container for the Public Certificate and Private Key.

This step combines the new signed Certificate and Private Key into a PKCS12 keystore. This is a container mechanism for securely storing certificates and private keys which can be put on the file system of each service. The service will read this keystore and obtain its certificate. When the service connects to another system, it will present this certificate to the other system for verification.

```
# Add the CA and Service Certificates to a P12 container without a password
openssl pkcs12 -export -nodes -out keystore.p12 -inkey service.key -in service.crt -certfile
rootCA.crt -name service -passout pass:
```

The certificate is given an alias when imported into the PKCS12 container. This alias can be the same across all of the Enterprise OMS services. It is best to use the name service in order to make it easier to configure each of the services. As each Enterprise OMS service requires its own keystore,

with its own certificates, the aliases will never conflict.

The keystore can be password-protected by omitting the `-passout pass:` parameter. A password prompt will ask for the password to use. It is important to understand that if the keystore is password-protected, then that password will need to be configured for each of the Enterprise OMS services so that it can read the keystore.

9.4. Add Root CA to Trust Store

Finally, the Root Certificate Authority is required for this entire process to work. Nothing will trust the manually created Certificate Authority until it is added to a Trust Store. By default, that Trust Store for the JVM is located at `JAVA_HOME/lib/security/cacerts`. This will import the Root CA into that Trust Store so that the Service can validate any certificates sent to it. The JVM will not work without the default trusted Certificates, so the Root CA must be added to the existing Trust Store, rather than creating a new one.

```
keytool -import -trustcacerts -keystore "%JAVA_HOME%\lib\security\cacerts" -storepass changeit -alias WorkModelCA -file rootCA.crt
```

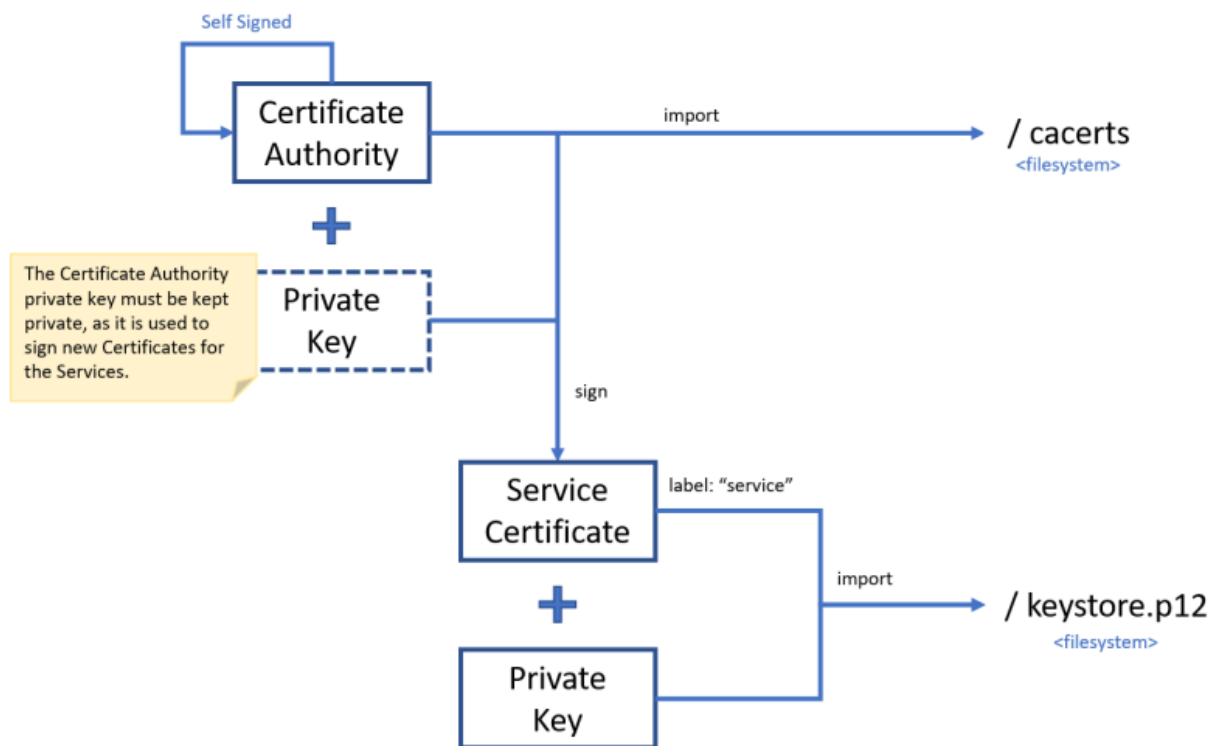


Figure 10. Public Key Infrastructure Process

9.5. Additional Certificate Operations

Create Self-Signed CA Script

Run Once

```
openssl genrsa -out rootCA.key 2048
openssl req -x509 -new -nodes -key rootCA.key -sha256 -days 3650 -out rootCA.crt -subj
"/C=US/ST=FL/O=GE Digital/CN=Work Model CA"
keytool -import -trustcacerts -keystore "%JAVA_HOME%\lib\security\cacerts" -storepass changeit -alias
WorkModelCA -file rootCA.crt
```

Create Service Certificate Script

Run for Each Service. Move the keystore.p12 into each Service's filesystem, then repeat.

```
openssl genrsa -out service.key 2048
openssl req -new -sha256 -key service.key -subj "/C=US/ST=CA/O=GE Digital/CN=localhost" -out
service.csr
openssl x509 -req -in service.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out service.crt
-days 3650 -sha256
openssl pkcs12 -export -nodes -out keystore.p12 -inkey service.key -in service.crt -certfile
rootCA.crt -name service -passout pass:
```

9.5.1. Common Keytool and Openssl Commands

Export Certificate from CACERTS

```
keytool -exportcert -alias WorkModelCA -keystore cacerts -storepass changeit -rfc -file capem
```

Export Cert from PKCS12 Container

```
keytool -exportcert -alias service -keystore postgres.p12 -storepass password -rfc -file postgres.crt
keytool -exportcert -alias service -keystore keystore.p12 -storepass password -rfc -file server.crt
```

Export Server Key from PKCS12 Container

```
openssl pkcs12 -in keystore.p12 -nodes -nocerts -out server.key
```

Export Server Key from PKCS12 to PKCS#8 DER format

```
openssl pkcs8 -topk8 -inform PEM -in "postgres.key" -outform DER -out "postgres.key.der" -v1 PBE-MD5-
DES
```

Remove password from server key

```
openssl rsa -in server.key -out server-nopass.key
```

Powershell: Import Certificates

```
$secure = Read-Host -AsSecureString
Import-PfxCertificate -FilePath .\work-model-eventhandler\keystore.p12 -CertStoreLocation
Cert:\CurrentUser\My -Exportable -Password $secure
Import-Certificate -FilePath .\ca.pem -CertStoreLocation Cert:\CurrentUser\Root
```

10. Enterprise OMS Journaling and Archiving

The Enterprise OMS system supports writing operational events to Work Model journal files and then writing data from the journal files to the Enterprise OMS database and the ADMS Archive database.

Work Model Journal Files

Work Model journal files are written by the Work Order Archive Adapter application. Work Order Archive Adapter communicates for the events with the Work Model Notification service and creates the journal entries based on the received events, such as crew assignment updates.

Work Model journal files are created to facilitate the replication of Enterprise OMS database and archive Enterprise OMS history in the ADMS Archive database. Work Model journal files use the same journal format as all ADMS journals.

Work Model journals include the following journal message types:

- WMAddOrderJournalMessage
- WMUpdateOrderJournalMessage
- WMDeleteOrderJournalMessage
- WMAddTaskJournalMessage
- WMUpdateTaskJournalMessage
- WMDeleteTaskJournalMessage
- WMAddDeviceOperationJournalMessage
- WMUpdateDeviceOperationJournalMessage
- WMDeleteDeviceOperationJournalMessage
- WMAddCrewJournalMessage
- WMUpdateCrewJournalMessage
- WMDeleteCrewJournalMessage

Writing Data from Work Model Journals to the Enterprise OMS Database

The Work Order Database Sync application reads Work Model journal files and writes the data to the Enterprise OMS database.

Writing Data from Work Model Journals to the ADMS Archive Database

The ADMS Archive application reads Work Model journal files and writes the data to the ADMS Archive database.

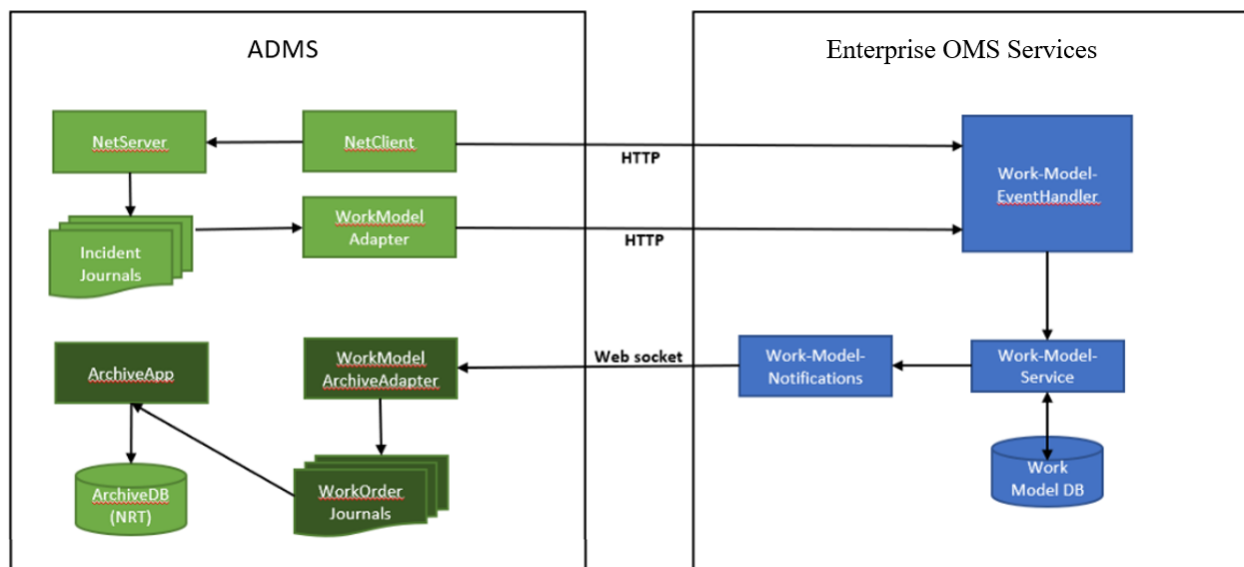


Figure 11. Archiving Enterprise OMS Data

11. Enterprise OMS Organizations Definition

This section provides information about the organizations function in the Enterprise OMS and Outage Management System. An "organization" is a group of people who perform a work function, such as the Dispatch (Delta) organization performs order-dispatching in the Delta area, Construction-Maintenance (East) performs maintenance works in the East area, Hazard Management manages all emergency orders in the system, and Forestry (East) manages vegetation work in the East area. Areas (polygon groups) are defined in the customer file. An "organization type" is a category of organization, such as Dispatch, Planning, Storm Teams, or Forestry. An organization's jurisdiction might be divided geographically across the service territory, or it might be divided by some other criteria such as the severity of a problem.

An organization has some similarities and differences to an area of responsibility (AOR).

Organization	AOR
An organization grants authority to see or manipulate groups of orders.	An AOR primarily grants authority to view or manipulate certain network objects/devices.
An organization is assigned to orders at run time and can be changed, if desired	An AOR is assigned to objects/devices at model time. It might be copied to an order, but it never changes.
Upon initial creation of an order, an organization can be automatically assigned based on predefined criteria.	
An order can be transferred to a different organization, if desired.	
For manually created orders (for example, follow-up orders), an initial organization can be manually selected at creation time	

Note

The terms AREA and AOR (area of responsibility) are often used interchangeably.

To define organizations for the OMS and Enterprise OMS systems:

1. Determine the organization types and organizations based on your operations requirements.
2. Decide how each organization type's jurisdictions will be determined. For example, they might be determined geographically.
3. Decide the criteria for assigning organizations to Orders based on Incident, Order type, and Call type.
4. Define a default "fall-back" organization name. In the event that OMS cannot determine an organization by other normal means, this name would be used.
5. Use the Coded Translation Editor to define your organizations (WorkOrganizations) and organization types (WorkOrganizationTypes).
6. Use the Permit application to define your organizations as AREAs and MODEs. You will also define the user types and users for organization membership.

7. Use the OMS Server Configuration Editor to define the criteria for assigning organizations to orders.
8. For organization types that will assign based on geography, define ESRI shape files using a GIS tool.
9. Use the Customer List Compiler to define and import any ESRI shape files into the customer model file.

11.1. Determining Organization Types and Organizations

Based on the various groups of people who will use the system, you must determine organization types and the individual organizations within them.

The following examples of geographically and non-geographically based organization types and the associated organizations are configured in the Fantasy Island demo model:

- **Dispatch:** Dispatcher organization type, geographically-based. Defined organizations: Dispatch Organization, Dispatch (Beaches), Dispatch (Highlands), Dispatch (Plains), Dispatch (Delta).
- **Planning:** Planning group organization type, non-geographically-based. Only one organization is defined: Storm Team.
- **Storm Teams:** Storm desk organization type, non-geographically-based. Only one organization is defined: Planning.
- **Construction-Maintenance:** Construction and maintenance organization type, geographically-based. Defined organizations: C-M (East), C-M (West), C-M (North), C-M (South).
- **Forestry:** Vegetation and forestry organization type, geographically-based. Defined organizations: Forestry (East), Forestry (West), Forestry (North), Forestry (South).
- **Hazard Management:** Hazard and emergency management organization type, non-geographically-based. Only one organization is defined: Hazard Management.
- **System Operations:** System operations organization type, geographically-based. Defined organizations: SysOps (Beaches), SysOps (Highlands), SysOps (Plains), SysOps (Delta).
- **Power Quality:** Power quality operations organization type, geographically-based. Defined organizations: Power Quality (East), Power Quality (West), Power Quality (North), Power Quality (South).
- **Data Quality:** Data quality operations organization type, non-geographically-based. Only one organization is defined: Data Quality.
- **Unassociated Calls:** Unassociated call operations organization type, non-geographically-based. Only one organization is defined: Unassociated Calls.

Note

The ID of an organization will also be used for an AREA in the Permit database. Therefore, the

organization ID (for example, STORM) must be limited to 8 characters in length.

11.2. Defining the Criteria for Assigning Organizations

Based on which type of organization would be assigned for each combination of incident/order type and call type, you can examine the incident types and call types in the Coded Translation Editor using the OutageIncidentType, NonoutageType, and CallType coded translation items.

For more information about the Coded Translation Editor, refer to the *Coded Translation Editor User Guide*.

Note

The same identifier might exist as both an OutageIncidentType and a NonoutageType. This is acceptable because you can specify whether your configuration applies to the outage or nonoutage type in the organization configuration in the OMS Server Configuration Editor.

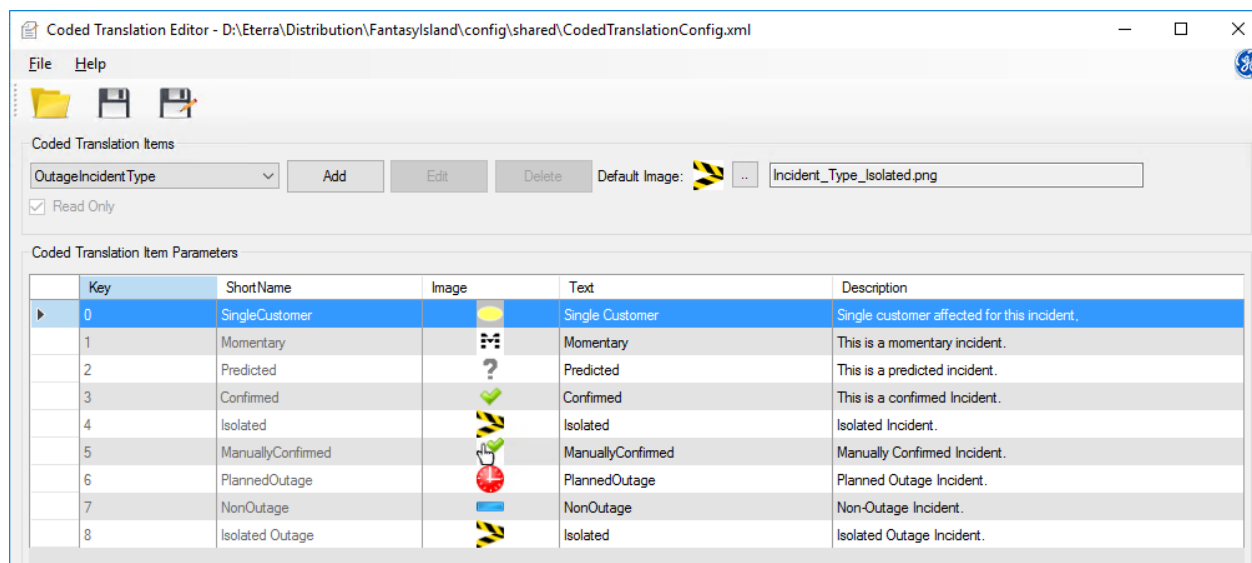


Figure 12. OutageIncidentType Coded Translation Item

Coded Translation Items

NonoutageType ▼ Add Edit Delete Default Image:  FindHeaderLine.bmp

☐ Read Only

Coded Translation Item Parameters

Key	ShortName	Image	Text	Description
0	Outage		Outage	Outage
1	Isolated		Isolated	Isolated Incident.
2	StreetLight		Street Light	Street Light
3	Hazard		Hazard	Hazard
4	Momentary		Momentary	This is a momentary non outage incident.
5	TrafficLight		Traffic Light	Traffic Light
6	PoleMaintenance		Pole Maintenance	Pole Maintenance
7	WireDown		Wire Down	Wire Down
8	IS Trouble		IS Trouble	IS Trouble
9	Service Call		Service Call	Service Call
10	Maintenance		Maintenance	Maintenance incident

Figure 13. NonoutageType Coded Translation Item

Coded Translation Items

CallType ▼ Add Edit Delete Default Image:  Calls.png

☐ Read Only

Coded Translation Item Parameters

Key	ShortName	Image	Text	Description
0	None		None	None
1	Lights Out		Lights Out	Lights Out
2	Area Out		Area Out	Area Out
3	Part Power		Part Power	Part Power
4	Dim/Bright Lights		Dim/Bright Lights	Dim/Bright Lights
5	Dim Lights		Dim Lights	Dim Lights
6	Blinking Lights		Blinking Lights	Blinking Lights
7	SL Out		SL Out	SL Out
8	SL Day Burn		SL Day Burn	SL Day Burn
9	In Service Call		In Service Call	In Service Call
10	Misc		Misc	Misc
11	Inv		Inv	Inv
12	Hazard Call		Hazard Call	Hazard Call
13	Emergency/Injury		Emergency/Injury	Emergency/Injury
14	Still Out		Still Out	Still Out
15	>911		>911	>911

Figure 14. CallType Coded Translation Item

11.3. Define a Default "Fall-Back" Organization Name

If OMS cannot determine an organization using configurations in the OMS Client Configuration Editor, the default organization name is used. For example, there might be a separate organization name that denotes "unassigned", which only a dispatch supervisor accesses.

The default organization name is configured using the DISPATCH_ORGANIZATION_AOR_NAME parameter in the NetServerOmsConfig.txt file. For example:

```
DISPATCH_ORGANIZATION_AOR_NAME = DISPORG
```

11.4. Defining Organization Types and Organizations

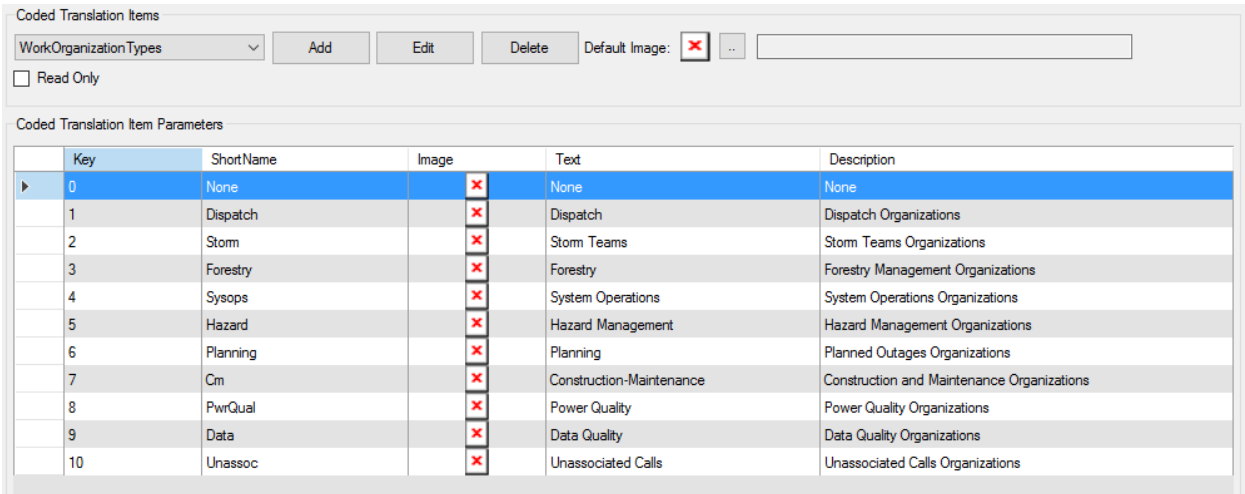
After you have determined the organization types and organization have determined the criteria for assigning to each, you can define them in the product using coded translation items.

Organization Types

Your organization types must be defined in the WorkOrganizationTypes coded translation item.

Note

The ShortName is the identifier used by OMS. This value is shown on the user interface (for example, in list boxes and column fields). Enter a meaningful short name, preferably without spaces.



Key	ShortName	Image	Text	Description
0	None		None	None
1	Dispatch		Dispatch	Dispatch Organizations
2	Storm		Storm Teams	Storm Teams Organizations
3	Forestry		Forestry	Forestry Management Organizations
4	Sysops		System Operations	System Operations Organizations
5	Hazard		Hazard Management	Hazard Management Organizations
6	Planning		Planning	Planned Outages Organizations
7	Cm		Construction-Maintenance	Construction and Maintenance Organizations
8	PwrQual		Power Quality	Power Quality Organizations
9	Data		Data Quality	Data Quality Organizations
10	Unassoc		Unassociated Calls	Unassociated Calls Organizations

Figure 15. WorkOrganizationTypes Coded Translation Item

Organizations

Your organizations must be defined in the WorkOrganizations coded translation item.

Note

The WorkOrganizations short names must be upper case, without space, and 8 characters or less in length, as they will match an AREA in the Permit database.

Coded Translation Items

WorkOrganizations ▼ Add Edit Delete Default Image: ...

☐ Read Only

Coded Translation Item Parameters

Key	ShortName	Image	Text	Description
0	Unassigned		Unassigned	No organization assigned
1	DISPORG		Dispatch Organization	Default dispatch organization
2	DSP_BCHS		Dispatch (Beaches)	Dispatch (Beaches) organization
3	DSP_HIGH		Dispatch (Highlands)	Dispatch (Highlands) organization
4	DSP_PLNS		Dispatch (Plains)	Dispatch (Plains) organization
5	DSP_DLTA		Dispatch (Delta)	Dispatch (Delta) organization
6	STORM		Storm Team	Storm Team organization
7	FOR_EAST		Forestry (East)	Forestry Organization East
8	FOR_WEST		Forestry (West)	Forestry Organization West
9	FOR_NRTH		Forestry (North)	Forestry Organization North
10	FOR_STH		Forestry (South)	Forestry Organization South
11	CM_EAST		C-M (East)	Construction-Maintenance Organization East
12	CM_WEST		C-M (West)	Construction-Maintenance Organization West
13	CM_NRTH		C-M (North)	Construction-Maintenance Organization North
14	CM_STH		C-M (South)	Construction-Maintenance Organization South
15	PQ_EAST		Power Quality (East)	Power Quality Organization East
16	PQ_WEST		Power Quality (West)	Power Quality Organization West
17	PQ_NRTH		Power Quality (North)	Power Quality Organization North
18	PQ_STH		Power Quality (South)	Power Quality Organization South
19	SO_BCHS		SysOps (Beaches)	SysOps (Beaches) organization
20	SO_HIGH		SysOps (Highlands)	SysOps (Highlands) organization

Move Up Move Down Add Delete

Figure 16. WorkOrganizations Coded Translation Item

11.5. Using the Permit Application to Define Organizations

Note

Complete Permit modeling information is out of scope of this document. For more details, refer to the *Habitat PERMIT User Guide*.

When using the hierarchical mode, permissions for the areas referenced in the hierarchical model can only be managed from the Permission Management plug-in. If areas referenced in the hierarchical model are also referenced in MODEs, mixing hierarchical permissions and normal Permit MODE management results in permissions inconsistency.

If you intend to use both MODE switching (the design for Enterprise OMS organizations) and the Permission Management plug-in, make sure that you define a compatible model. If you have inconsistencies, contact your GE project team.

It is assumed that you are not creating a new Permit model from scratch at this time (doing so is beyond the scope of these instructions). It is also assumed you have already defined a base MODE that grants the minimum rights (Applications and AREAs) to operate the system. (In the Fantasy

Island demonstration model, the base MODE is called DMSTMPLT.)

This Permit modeling task consists of modeling AREAs, MODEs, USERTYPs, and USERS. It is suggested that you do Permit modeling on a development system that contains the same Permit model that will be used on the production system. After the Permit modeling you can make a Permit save case and draw that case into production.

11.5.1. Adding AREAs

Note

The terms AREA and AOR (area of responsibility) are often used interchangeably.

Note

The AREA IDs must be 8 characters or less in length.

To add an AREA:

1. In the Browser client, click HABITAT Applications > Permit > Mode Management.
 2. On the Analyst Displays menu, click Mode Modeling > Area Definitions.
- The AREA_DEFINITIONS display appears.
3. Add an AREA corresponding to each of your new Enterprise OMS organizations.
- (Use the +/- button right-click menu to add or remove areas.)
4. Fill in IDs and Descriptions for organization areas.
 5. Add an AREA corresponding to the default "fall-back" organization name (that is defined by the OMS parameter, DISPATCH_ORGANIZATION_AOR_NAME).

PERMIT

Area Definitions

Import Applied: 07-Sep-2020 07:50:49

Validate Model

Apply Model

Import HMODE

Area ID		Critical Permissions			
		Read	Write	Execute	Audible
+/-	302 DSP_CTL	☐	☐	☐	☐
Description: OMS Dispatch area					
+/-	303 DSP_STH	☐	☐	☐	☐
Description: OMS Dispatch area					
+/-	305 DISPOG	☐	☐	☐	☐
Description:					

	307	PLANNING	☐	☐	☐	☐
		Description: Planned outages organization				
	308	SERVICEA	☐	☐	☐	☐
		Description:				
	309	SERVICEB	☐	☐	☐	☐

6. Ignore the settings for "Critical Permissions".

11.5.2. Adding MODEs

Note

The MODE IDs you will add do not necessarily have to exactly match the ID of the corresponding AREA. However, it is recommended to use names that are associated with AREA names to easily understand the matching AREA.

In a sense, the MODE is the organization. You could say the MODE packages the permissions for the organization.

To add a MODE:

1. In the Browser client, click HABITAT Applications > Permit > Mode Management.
2. On the Analyst Displays menu, click Mode Modeling > Mode Definitions.

The MODE_DEFINITIONS display appears.

3. Add or remove MODEs. (Use the +/- button right-click menu to add or remove Modes.)
4. Fill in IDs and Descriptions (as desired).
5. Add a MODE corresponding to each of your Enterprise OMS organizations. (You do not necessarily have to model a MODE for the DISPATCH_ORGANIZATION_AOR_NAME.)

PERMIT Mode Definitions

Validate Model Apply Model

Sort by Mode ID ▾

+/-	83	ID: VEGSOUTH	Navigate
Description:		OMS Vegetation management	
+/-	84	ID: DSP_NEA	Navigate
Description:		OMS dispatch area	
+/-	85	ID: DSP_WST	Navigate
Description:		OMS dispatch area	

11.5.3. Granting AREA Access to MODEs

Note

The terms AREA and AOR (area of responsibility) are often used interchangeably.

Each new MODE must be granted enough AREA access so that an organization member can do all necessary operations, such as see and modify orders, and operate devices.

As a minimum, each MODE would need access to:

- The AREA bearing the MODE's organization name (for example, MODE DSP_WST needs access to AREA DSP_WST).
- Any AREA that might belong to a network object (device), station, or service center that falls within this MODE's (organization's) jurisdiction. For example, mode DSP_WST would have access to AREAs:
 - WSTA, WSTB, WSTC, 3LIONSSC as they represent service centers.
 - 3LIONS and COHO (among others) as they represent stations.
 - 3LIONSNE, 3LIONSNW, 3LIONSSE, 3LIONSSW, COHONE, COHONW, COHOSE, COHOSW (among others) as they represent AREAs that might be assigned to network objects (devices) in the pertinent region.
- Any AREA that represents a *functional* AOR that is required for a member of this organization to

perform their job. See the *ADMS Software Installation and Maintenance Guide* for detailed information. Examples include:

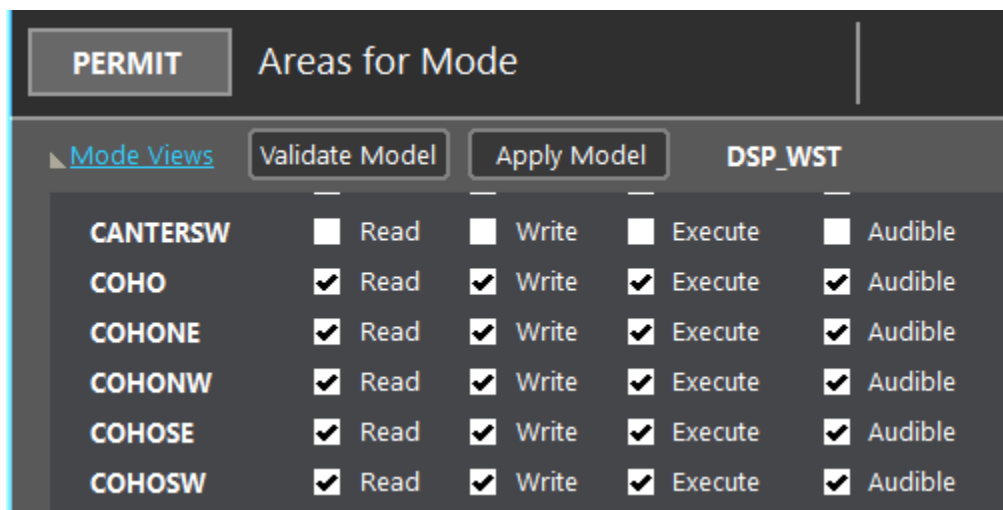
- SWITCHOP
 - DISPATCH
 - OUTAGE
 - TAGGING
 - EDISON
- Any default AREAs that may be required, such as the DISPATCH_ORGANIZATION_AOR_NAME value or DEFPAREA.
 - As required, any AREA that corresponds to an organizational hierarchy (note that this is a different, unrelated use of the term "organization") level that the organization member may need. (It is possible that only the service center is needed.)

To grant permissions to an AREA:

1. In the Browser client, click HABITAT Applications > Permit > Mode Management.
2. On the Analyst Displays menu, click Mode Modeling > Mode Directory.

The MODE_DIRECTORY display appears.

3. Locate your new mode in the list, right-click on it, and select Mode Areas.
4. Set Read, Write, Execute, and Audible permissions for any AREA your organization membership requires, as decided above.



PERMIT Areas for Mode				
Mode Views	Validate Model	Apply Model	DSP_WST	
NEAD	☐ Read	☐ Write	☐ Execute	☐ Audible
WST	☒ Read	☒ Write	☒ Execute	☒ Audible
WSTA	☒ Read	☒ Write	☒ Execute	☒ Audible
WSTB	☒ Read	☒ Write	☒ Execute	☒ Audible
WSTC	☒ Read	☒ Write	☒ Execute	☒ Audible

PERMIT Areas for Mode				
Mode Views	Validate Model	Apply Model	DSP_WST	
EDISON	☒ Read	☒ Write	☒ Execute	☒ Audible
ARBOR	☐ Read	☐ Write	☐ Execute	☐ Audible
OUTAGE	☒ Read	☒ Write	☒ Execute	☒ Audible
STORMDSK	☐ Read	☐ Write	☐ Execute	☐ Audible
VEGEAST	☐ Read	☐ Write	☐ Execute	☐ Audible
VEGWEST	☐ Read	☐ Write	☐ Execute	☐ Audible
VEGNORTH	☐ Read	☐ Write	☐ Execute	☐ Audible
VEGSOUTH	☐ Read	☐ Write	☐ Execute	☐ Audible
DSP_NEA	☐ Read	☐ Write	☐ Execute	☐ Audible
DSP_WST	☒ Read	☒ Write	☒ Execute	☒ Audible
DSP_CTL	☐ Read	☐ Write	☐ Execute	☐ Audible

Special Note for the PLANNING AREA

The demo Fantasy Island model contains a special example for the PLANNING AREA. The Dispatch organization members have the Read access to PLANNING by default, but not Write access. This is done by granting only Read to the PLANNING AREA for all the Dispatch MODES (for example, DSP_WST), as shown below.

+ image::image39.png[]

The result is that DSP_WST members (or any other MODE set up this way) can see PLANNING organization Orders on their lists but cannot modify them.

11.5.4. Adding User Types

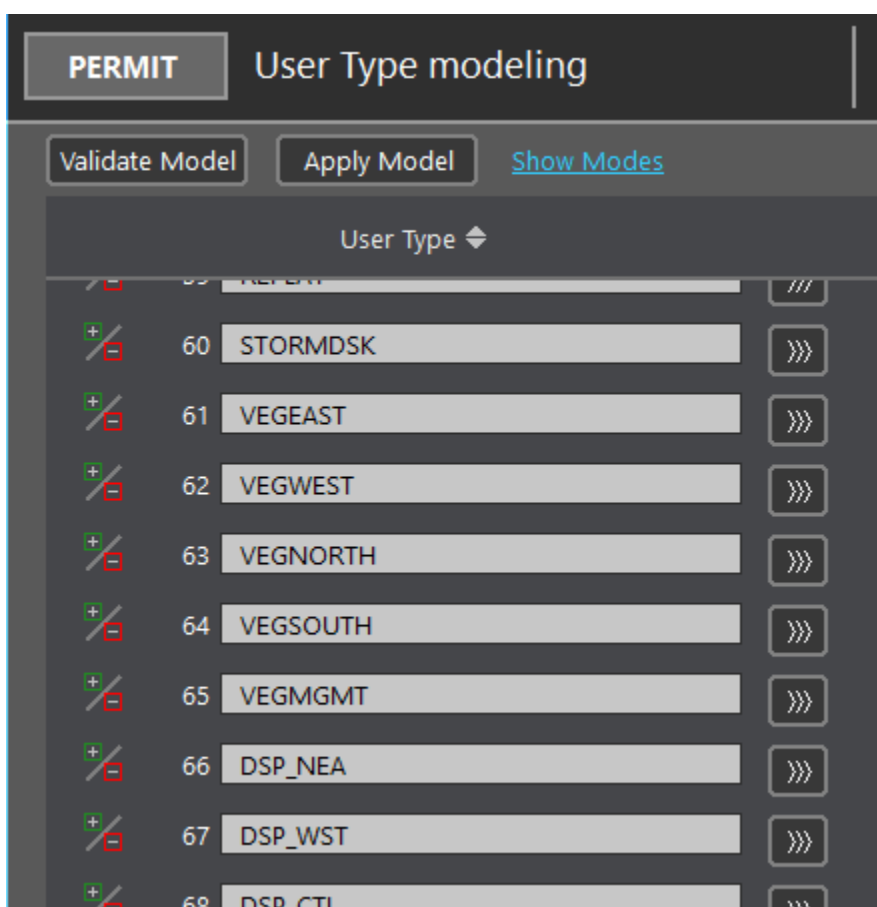
Typically, you need to add a user type for each specific new organization you have modeled (for example, DSP_WST, DSP_NEA, and STORMDSK) except for the default organization name. The ID of a user type does not need to match anything, but you should choose names that indicate the purpose.

To add a user type:

1. In the Browser client, click HABITAT Applications > Permit > Mode Management.
2. On the Analyst Displays menu, click User Modeling > User Type Modeling.

The USRTYP_MODEL_MINI, PERMIT display appears.

3. Add or remove user types that are members of one organization (for example, DSP_CTL). (Use the +/- button right-click menu options to add or remove users).



4. Add user types for users that are members of more than one organization.

For example, a DSP_ALL user type might have access to multiple organizations: DSP_CTL, DSP_WST, DSP_NEA, and DSP_STH. A SVCSUPV user type might have access to both SERVICEA and SERVICEB organizations. A VEGMGMT type might have access to all vegetation management organizations.



11.5.5. Assigning MODEs to User Types

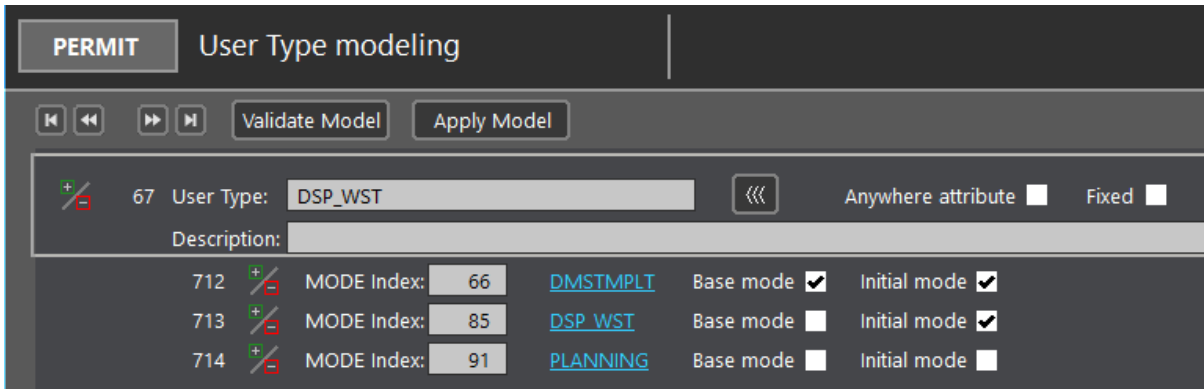
You need to assign organization MODEs to user types to make the user types an organization member type.

To add a user type:

1. In the Browser client, click HABITAT APPLICATIONS > Permit > Mode Management.
2. On the Analyst Displays menu, click User Modeling > User Type Modeling.

The USRTYP_MODEL_MINI, PERMIT display appears.

3. Find the user type entry (USRTYP) in the list and click the chevron (☐) symbol to expand the definition.



Note

You can press the Show Modes button on the upper right to see the index of all modes.

4. Ignore the Anywhere and Fixed flags for purposes of this procedure.
5. Add or remove user modes (UMODE) as needed using the +/- button (right click), and fill in the index of the desired MODE.

You must include:

- The base UMODE for system operation (for example, DMSTMPLT) as mentioned at the beginning of this procedure. Mark this UMODE as Base and Initial.

Note

A Base UMODE cannot be dropped in runtime by a user, and there can be only one. Non-base UMODEs can be dropped or added in runtime, and there can be more than one.

- The mode corresponding to the organization (for example, DSP_WST). Mark this UMODE as Initial.

Note

Special note for PLANNING – in our example we have added the ability for the dispatch user types to add the full PLANNING mode in runtime if they desire. But that will not be the default, because that UMODE is not marked as Initial.

Below is an example of a supervisory dispatch organization user type who has access to multiple organizations. Note that they could also add PLANNING or any one of the vegetation UMODEs in runtime if they want, but they are not Initial UMODEs.

PERMIT

User Type modeling

◀

◀◀

▶▶

▶

Validate Model

Apply Model

+

-

70

User Type: DSP_ALL

⏏

Anywhere attribute ☐

Description:

721	+ -	MODE Index: 66	DMSTMPLT	Base mode ☒	Initial mode ☒
722	+ -	MODE Index: 84	DSP_NEA	Base mode ☐	Initial mode ☒
723	+ -	MODE Index: 85	DSP_WST	Base mode ☐	Initial mode ☒
724	+ -	MODE Index: 86	DSP_CTL	Base mode ☐	Initial mode ☒
725	+ -	MODE Index: 87	DSP_STH	Base mode ☐	Initial mode ☒
726	+ -	MODE Index: 91	PLANNING	Base mode ☐	Initial mode ☐
727	+ -	MODE Index: 80	VEGEAST	Base mode ☐	Initial mode ☐
728	+ -	MODE Index: 81	VEGWEST	Base mode ☐	Initial mode ☐
729	+ -	MODE Index: 82	VEGNORTH	Base mode ☐	Initial mode ☐
730	+ -	MODE Index: 83	VEGSOUTH	Base mode ☐	Initial mode ☐

Below is an example of a vegetation management organization user type that initially has access only to their own organization but could add others in runtime if desired.

PERMIT

User Type modeling

⏮

⏪

⏩

⏭

Validate Model

Apply Model

+

-

62

User Type:

VEGWEST

⏮

Anywhere attribute

☐

Description:

OMS vegetation management assignment

689	+	-	MODE Index:	66	DMSTMPLT	Base mode	☒	Initial mode	☒
690	+	-	MODE Index:	81	VEGWEST	Base mode	☐	Initial mode	☒
691	+	-	MODE Index:	80	VEGEAST	Base mode	☐	Initial mode	☐
692	+	-	MODE Index:	82	VEGNORTH	Base mode	☐	Initial mode	☐
693	+	-	MODE Index:	83	VEGSOUTH	Base mode	☐	Initial mode	☐

11.5.6. Defining Users

Decide which user IDs will be members of which organizations.

To add a user:

1. In the Browser client, click HABITAT Applications > Permit > Mode Management.
2. On the Analyst Displays menu, click User Modeling > User Modeling.

The USRTYP_MODEL_MINI, PERMIT display appears.

3. Add or remove users (USER)

(Use the +/- button right-click menu to add or remove areas.).

4. Fill in the index of the desire user type.

PERMIT

User Modeling

Validate Model

Apply Model

	User ID	Disabled	USRTYP Index	
+	62	☐	68	DSP_CTL
+	63	☐	69	DSP_STH
+	64	☐	70	DSP_ALL
+	65	☐	71	PLANNING

Note

You can click the Show User Types button on the upper right to see the index of all types.

You can press the chevron symbol to expand the definition of the USER, but the definition is beyond the scope of this procedure. For more details, see the *Habitat PERMIT User Guide*.

11.5.7. Validating and Saving the Model

To validate and save the model:

1. In the Browser client, click HABITAT Applications > Permit > Mode Management.
2. On the Analyst Displays menu, click User Modeling > User Modeling.
The USRTYP_MODEL_MINI, PERMIT display appears.
3. Click Validate Model.
4. On the Permit Server Displays menu, click Permit Server Log.
5. Correct errors, as appropriate.
6. When the model is successfully validated, click Apply Model.
7. Make a Permit Both save case, so it can be transferred to a production system. For more details, refer to the *Habitat User Guide*.

11.6. Defining Criteria for Assigning Organizations

For more information about configuring criteria for assigning organizations to specific incident types, refer to the chapter "Organizations Tab" in the *Outage Management System Server Configuration Editor User Guide*. Use the criteria you defined in [Defining the Criteria for Assigning Organizations](#).

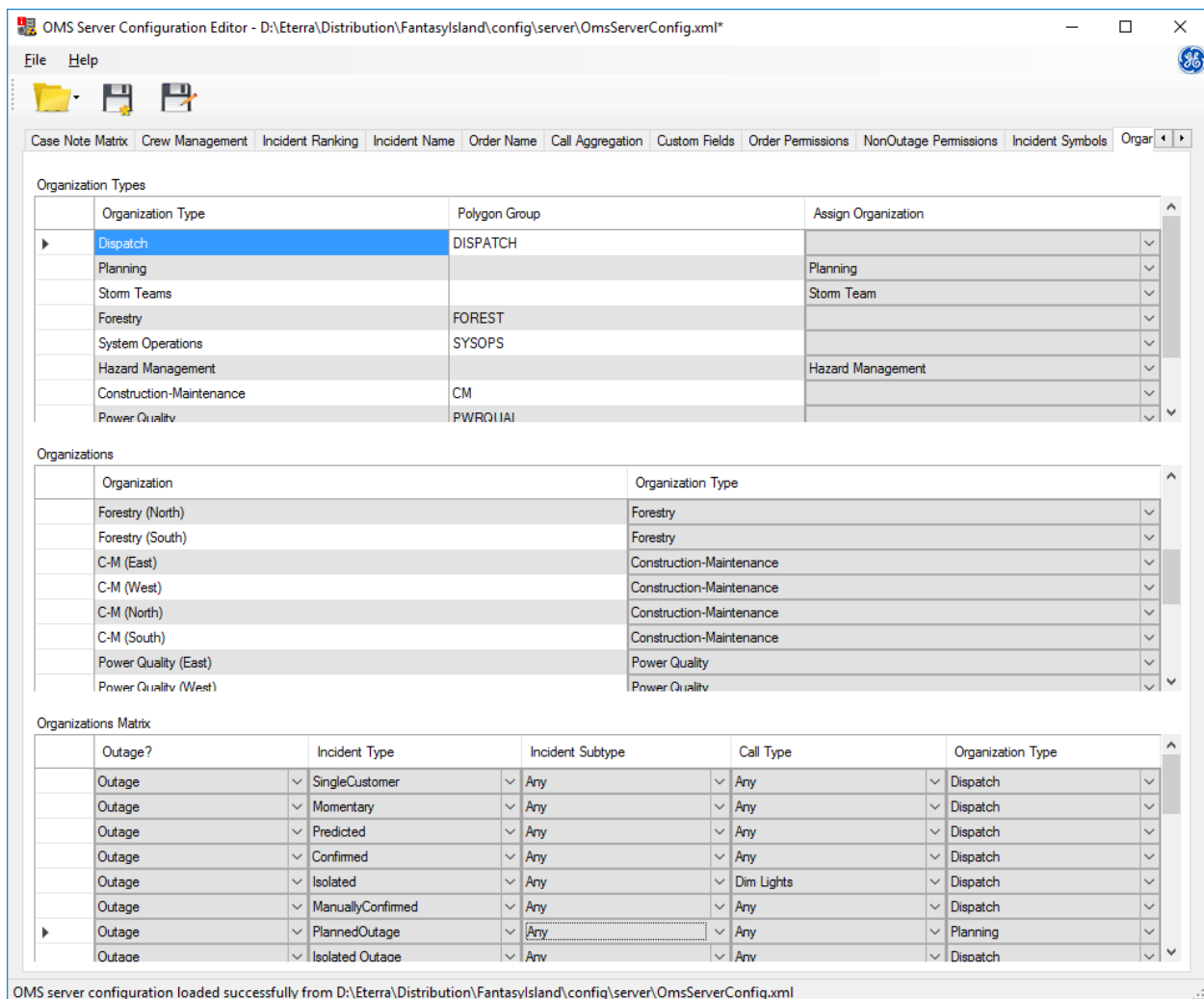


Figure 17. Defining Criteria for Assigning Organizations

11.7. Defining ESRI Shape Files

Note

ESRI shape files can be defined using a typical third-party GIS system. Complete instructions for defining ESRI shape files are beyond the scope of this procedure.

ESRI shape files hold the polygons which define the boundaries of organizations. These are simple multipolygon shape files with attributes.

The following figure shows the DISPORG (Dispatch) polygons. These polygons (features) subdivide the entire Fantasy Island footprint into four regions which correspond to the four dispatch organizations.

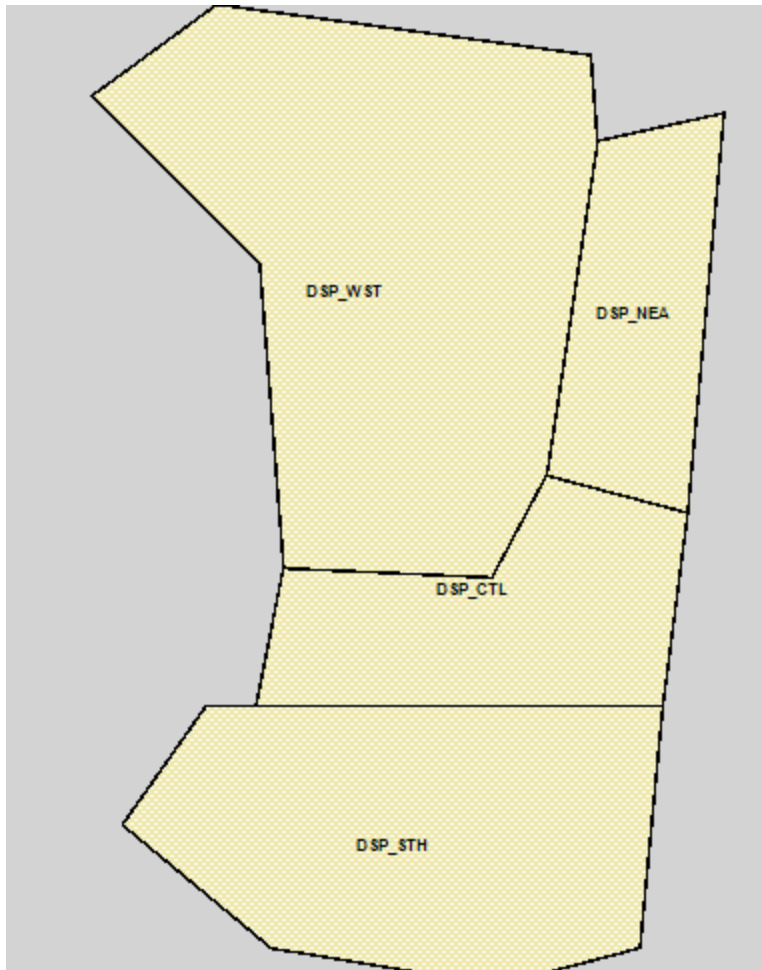


Figure 18. DISPOrg Polygons

Each polygon (feature) must include the following attributes:

- **TYPE_CODE:** This can be any name but should be the same for each feature (polygon) in the file.
- **ORG_ID:** This ID must match the organization names you defined in OMS (Organization) and Permit (AREA).
- **LABELTEXT:** This text must match the organization names you have defined in OMS (Organization) and Permit (AREA). It is recommended that this value matches ORG_ID.
- **ID:** This ID can be any value, which must be unique among features.

The following table shows the attributes for the DISPOrg shape file.

TYPE_CODE	ORG_ID	LABELTEXT	ID
DISP	DSP_CTL	DSP_CTL	1
DISP	DSP_NEA	DSP_NEA	2
DISP	DSP_STH	DSP_STH	3
DISP	DSP_WST	DSP_WST	4

The following files of the ESRI shape file set are required by ADMS:

- The .shp file (the main shape file)
- The .dbf file (the attributes database file)
- The .shx file (the index file)

Each file must have the same prefix.

11.8. Applying ESRI Shape Files to the Customer Model

To apply ESRI shape files to the customer model, use the Customer List Compiler. The following steps describe how to place your ESRI shape files in the appropriate location, define the files in the Customer List Compiler, and build the customer model file.

To import ESRI shape files:

1. Note the name and location of your customer data text file and schema file. (For the Fantasy Island sample, it is `FantasyIslandCustomers.txt` and `FantasyIsland_customerSchema.xml` respectively.)
2. Start the Customer List Compiler.
3. Open your customer schema file.

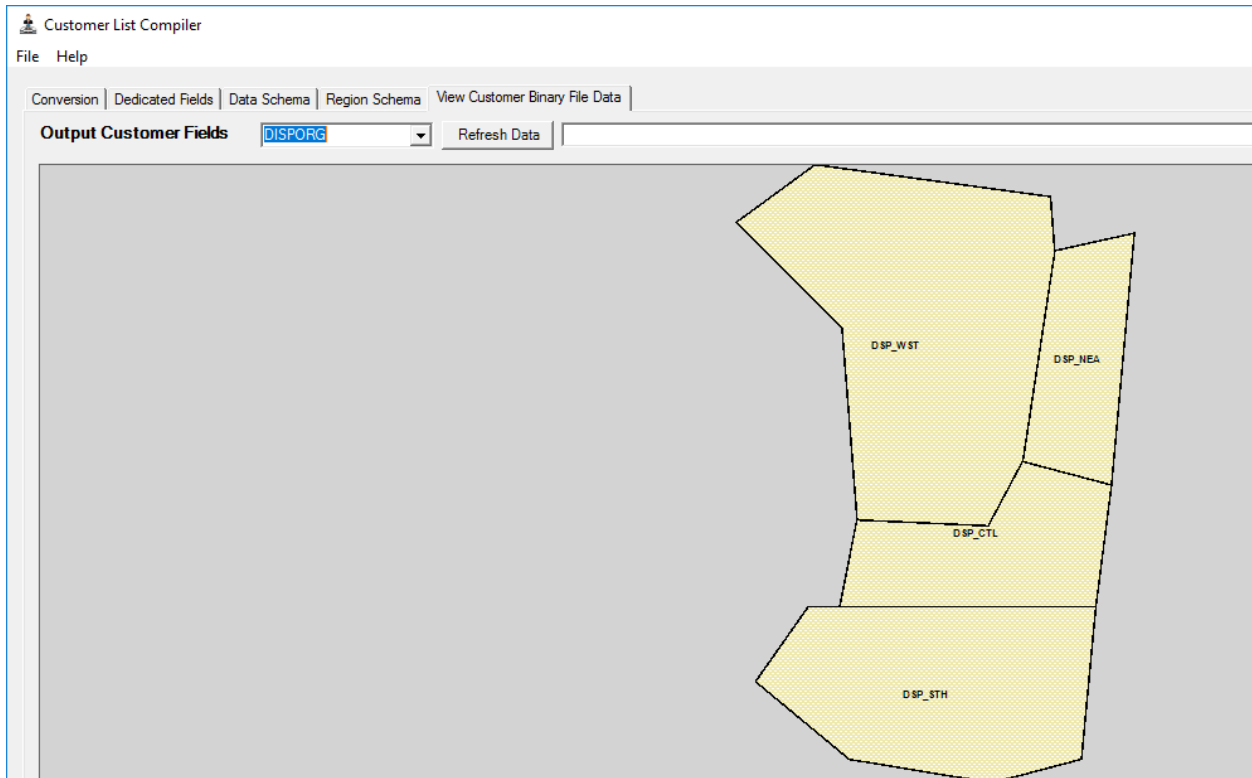
For more details, refer to the *Customer List Compiler User Guide*.

4. Select the Dedicated Fields tab. In the Polygon Groups section, enter the data corresponding to your ESRI shape file.

	Polygon Group Name	Shape File Location	Primary Label Column Name	PrimaryNameColumnName	Alternate Name Column Name
	SVCENTER	%IDMS_ROOT%\ToolsWorkspace\CustomerCompiler\FantasyIslandSvcCenters.shp	ORG_ID	ORG_ID	LABELTEXT
	VEGMGMT	%IDMS_ROOT%\ToolsWorkspace\CustomerCompiler\FIVegManagement.shp	ORG_ID	ORG_ID	LABELTEXT
▶	DISPORG	%IDMS_ROOT%\ToolsWorkspace\CustomerCompiler\FIDispatchOrgs.shp	ORG_ID	ORG_ID	LABELTEXT
	SVCORG	%IDMS_ROOT%\ToolsWorkspace\CustomerCompiler\FIServiceOrgs.shp	ORG_ID	ORG_ID	LABELTEXT

- Polygon Group Name: Enter the name you defined in the Organizations tab in the Outage Management System Server Configuration Editor.
 - Shape File Location: Enter the location where you will place the .shp, .dbf, and the .shx files.
 - Primary Label Column Name: Enter the ORG_ID as seen in the shape file attributes.
 - Primary Name Column Name: Enter the ORG_ID as seen in the shape file attributes.
 - Alternate Name Column Name: Enter the LABELTEXT as seen in the shape file attributes.
5. Save the schema.
 6. Place your .shp, .dbf, and the .shx files in the location you specified in the Shape File Location column.

7. On the Conversion tab, click Compile Customers and select your customer data text file.
8. Correct any errors and recompile as necessary.
9. Open your compiled customer model file.
10. Navigate to the View Customer Binary File Data tab. You should be able to select your new Polygon Group in Output Customer Fields, press Refresh Data, and see your polygons.



12. Configuration Files

12.1. WorkOrderAdapterConfig.xml

The Work Order Adapter reads OMS data from journal files, snapshots, and customer and crew models and then sends data updates to Enterprise OMS services through REST API. The Work Order Adapter communicates with the Work Model Event Handler service.

The Work Order Adapter uses the `WorkOrderAdapterConfig.xml` configuration file, which is located in the `%IDMS_ROOT%\config\interfaces` folder.

The `WorkOrderAdapterConfig.xml` configuration file includes the following parameters:

- **WorkOrderServiceUrl:** The Work Model Event Handler service REST API URL. For a high availability configuration, you can specify multiple URLs, separated by commas. The Work Order Adapter can identify and connect to the active Work Model Event Handler service.
- **WorkOrderTlsEnabled:** Specifies whether the HTTPS TLS security is enabled for the Work Model Event Handler service connection for web clients and web socket clients.
- **WorkModelTlsIgnoreRemoteNameMismatch:** Specifies whether the remote server certificate name mismatch is ignored. If set to True, the remote certificate name can be different from the URL host name that is used to connect to the server.
- **WorkOrderTlsSubjectName:** The certificate subject name that is used when searching for client certificates that will be sent to the Work Model Event Handler service.
- **EnableDebugLog:** Specifies whether the Work Order Adapter debug log is recorded.
- **LogFileName:** Path to the text file where Work Order Adapter events are recorded in plain text format.
- **MaxMessagesInLogFile:** The maximum number of entries in a log file (by default, 10000). When a log file is opened for write, the code creates a new file with header information. When an entry is written to the log and the existing file has the maximum number of entries contained in it, a new file is created. A new file is also opened for the first entry written after midnight.
- **PersistentDataFile:** The persistent data file records data about already processed incident and crew journals so that these journals are not processed again if the Work Order Adapter is restarted.
- **SupportedUpdateIncidentFields:** Incident parameters whose update triggers sending UpdateIncident messages from the Work Order Adapter to the Enterprise OMS system. By default, incident parameters INCIDENTTYPE, ORDERSTATUS, and ISDEVICEOFFRECORDOPEN are configured as SupportedUpdateIncidentFields.
- **NetServerOmsConfig:** The path to the NetServerOmsConfig.txt file. The OMS_AUTOCLEAR_MOMENTARY_INCIDENTS value in NetServerOmsConfig.txt defines whether restored momentary incidents are auto cleared after the time period specified by OMS_MOMENTARY_OUTAGE_DURATION. If the auto clearance is disabled, momentary orders

are created for restored momentary incidents after the OMS_MOMENTARY_OUTAGE_DURATION timeout.

- **OmsCustomerFile:** Path to the customer model file
- **OmsCallJournalFilePath:** Path to the folder with Call journals.
- **OmsArchiveJournalFilePath:** Path to the folder with Archive journals.
- **OmsIncidentJournalFilePath:** Path to the folder with Incident journals.
- **OmsCrewJournalFilePath:** Path to the folder with Crew journals.
- **OmsOrderJournalFilePath:** Path to the folder with Follow-Up Work Order journals.
- **CodedTranslationConfig:** Path to the coded translation configuration file.
- **OmsSnapshotFilePath:** Path to the folder with snapshots.
- **OmsCrewEmployeeFile:** Path to the crew model file.

12.2. DataCorrectionInterfaceConfig.xml

In DataCorrectionInterfaceConfig.xml (by default, stored in %IDMS_ROOT%\config\interfaces), you can modify the following settings based on deployment needs.

- **GraphQLPort:** The GraphQL service port number the Data Correction Interface application to run on.
- **WorkOrderServiceUrl:** The Work Model Event Handler service REST API URL to get canceled and closed orders. For a high availability configuration, you can specify multiple URLs, separated by commas. The Data Correction Interface application can identify and connect to the active Work Model Event Handler service.
- **SecureWS:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode).

If SecureWS is set to true, you must set the CertificateSubject parameter value to the "Subject" (name) of the TLS certificate. This certificate must also be stored in the Trusted Root Certification Authorities folder of the certificate store. You must also change the Data Correction Interface's URI value to HTTPS in the BaseUri parameter in OmsDisplayPluginConfig.xml. For example:

```
<BaseUri>https://localhost:50001/dcgraphql</BaseUri>  
<OperationName>dataCorrectionData</OperationName>
```

- **CertificateSubject:** The TLS certificate subject (name).
- **WorkOrderTlsEnabled:** Specifies whether the HTTPS TLS security is enabled for the Work Model Event Handler Service connection for web clients and web socket clients.
- **WorkModelTlsIgnoreRemoteNameMismatch:** Specifies whether the remote server certificate name mismatch is ignored. If set to True, the remote server certificate name can be different from the URL host name that is used to connect to the server.

- **WorkOrderTlsSubjectName:** Specifies the certificate subject name that is used when searching for client certificates to send to the Work Model Event Handler Service.
- **DataCorrectionIncidentsHostURL:** The Net Data Server web API URL to get canceled and cleared incidents.
- **EnableDebugLog:** Specifies whether the detailed logging is enabled.
- **LogFileName:** Full path to the file where the Data Correction Interface log is recorded.
- **MaxMessagesInLogFile:** The maximum number of entries in the log file. When an entry is written to the log file and the existing file reaches the maximum number of entries, a new log file is created.